

**Differential Pressure and
Pressure Transmitters**

**EJ□110□, EJ□120□,
EJ□130□, EJ□310□,
EJ□430□, and EJ□440□**



IM 01C25B01-01E



Differential Pressure and Pressure Transmitters

EJ□110□, EJ□120□, EJ□130□,
EJ□310□, EJ□430□, and EJ□440□

IM 01C25B01-01E 28th Edition

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Revision Information		

When using the transmitters in a Safety Instrumented System application, refer to the Functional Safety Manual (Document No.: TI 01C25A05-01EN or TI 01C25A05-21EN for option code SLT) and follow the instructions and procedures described there. The document can be downloaded from the website of Yokogawa. (Website address: <https://www.yokogawa.com/solutions/products-platforms/field-instruments/>)

In order to satisfy the requirement of Safety Instrumented System, executing parameters setting is required. Please refer to the clause of "Setting Parameters" for setting range in the following manuals for the actual operation.

BRAIN: IM 01C25T03-01E
HART: IM 01C25T01-06EN

Please also refer to the contact output setting in the same clause. After installing the transmitter, confirm that the range and unit is set correctly. Calibration of the transmitters shall be done after completing the range setting.

1. Introduction

Thank you for purchasing the DPharp Differential Pressure and pressure transmitter.

Your transmitter was precisely calibrated at the factory before shipment. To ensure both safety and efficiency, please read this manual carefully before you operate the instrument.



NOTE

This manual describes the hardware configurations of the transmitters listed in below. For information on the software configuration and operation, please refer to either IM 01C25T03-01E for the BRAIN communication type, or IM 01C25T01-06EN for the HART communication type.

For FOUNDATION Fieldbus protocol type, please refer to IM 01C25T02-01E.

For PROFIBUS PA protocol type, please refer to IM 01C25T04-01EN.

For the specifications, external dimensions, and model, suffix and option codes of each product, please refer to the General Specifications sheet as listed in the Chapter 9.

Model	Style code
EJX110A	S3
EJX120A	S1
EJX130A	S2
EJX310A	S2
EJX430A	S2
EJX440A	S2
EJA110E	S1, S2
EJA120E	S1, S2
EJA130E	S1, S2
EJA310E	S1, S2
EJA430E	S1, S2
EJA440E	S1, S2

To ensure correct use of this instrument, read both the hardware and software manuals thoroughly before use.



WARNING

When using the transmitters in a Safety Instrumented System application, refer to the Functional Safety Manual (Document No.: TI 01C25A05-01EN or TI 01C25A05-21EN for option code SLT) and follow the instructions and procedures described there. The document can be downloaded from the website of Yokogawa. (Website address:

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BRAIN: IM 01C25T03-01E

HART: IM 01C25T01-06EN

Please also refer to the contact output setting in the same clause. After installing the transmitter, confirm that the range and unit is set correctly. Calibration of the transmitters shall be done after completing the range setting.



NOTE

When describing the model name like EJ□110□, it shows the applicability for both EJX110A and EJA110E. The same representations are used for the other models, too.



NOTE

Unless otherwise stated, the illustrations in this manual are of the EJ□110□ differential pressure transmitter. Users of the other models should bear in mind that certain features of their instrument will differ from those shown in the illustrations of the EJ□110□.

■ Regarding This Manual

- This manual and the identification tag attached on the packing box are essential parts of the product. Please keep them in a safe place for future reference.
- This manual should be provided to the end user.
- The contents of this manual are subject to change without prior notice.
- All rights reserved. No part of this manual may be reproduced in any form without Yokogawa's written permission.
- Yokogawa makes no warranty of any kind with regard to this manual, including, but not limited to, implied warranty of merchantability and fitness for a particular purpose.
- If any question arises or errors are found, or if any information is missing from this manual, please inform the nearest Yokogawa sales office.
- The specifications covered by this manual are limited to those for the standard type under the specified model number break-down and do not cover custom-made instruments.
- Please note that changes in the specifications, construction, or component parts of the instrument may not immediately be reflected in this manual at the time of change, provided that postponement of revisions will not cause difficulty to the user from a functional or performance standpoint.
- Yokogawa assumes no responsibilities for this product except as stated in the warranty.
- If the customer or any third party is harmed by the use of this product, Yokogawa assumes no responsibility for any such harm owing to any defects in the product which were not predictable, or for any indirect damages.
- The following safety symbols are used in this manual:



WARNING

Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.



CAUTION

Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.



IMPORTANT

Indicates that operating the hardware or software in this manner may damage it or lead to system failure.



NOTE

Draws attention to information essential for understanding the operation and features.

— Direct current

⊥ Functional grounding terminal

⚠ Caution

This symbol indicates that the operator must refer to an explanation in the user's manual in order to avoid the risk of injury or death of personnel or damage to the instrument.

■ Trademarks

- 'DPharp', 'EJX', 'EJA', 'FieldMate' and 'BRAIN TERMINAL' are registered trademarks or trademarks of Yokogawa Electric Corporation. Company names and product names used in this material are registered trademarks or trademarks of their respective owners.
- In this manual, trademarks or registered trademarks are not marked with TM or ®.

1.1 Safe Use of This Product

For the safety of the operator and to protect the instrument and the system, please be sure to follow this manual's safety instructions when handling this instrument. If these instructions are not heeded, the protection provided by this instrument may be impaired. In this case, Yokogawa cannot guarantee that the instrument can be safely operated. Please pay special attention to the following points:

(a) Installation



WARNING

- This instrument may only be installed by an engineer or technician who has an expert knowledge of this device. Operators are not allowed to carry out installation unless they meet this condition.
- With high process temperatures, care must be taken not to burn yourself by touching the instrument or its casing.
- Never loosen the process connector nuts when the instrument is installed in a process. This can lead to a sudden, explosive release of process fluids.
- When draining condensate from the pressure detector section, take appropriate precautions to prevent the inhalation of harmful vapors and the contact of toxic process fluids with the skin or eyes.
- When removing the instrument from a hazardous process, avoid contact with the fluid and the interior of the meter.
- All installation shall comply with local installation requirements and the local electrical code.

(b) Wiring



WARNING

- The instrument must be installed by an engineer or technician who has an expert knowledge of this instrument. Operators are not permitted to carry out wiring unless they meet this condition.
- Before connecting the power cables, please confirm that there is no current flowing through the cables and that the power supply to the instrument is switched off.

(c) Operation



WARNING

- Wait 5 min. after power is turned off, before opening the covers.
- Do not open the cover in wet weather or humid environment. If the cover is opened, stated enclosure protection is not applicable.

(d) Maintenance



WARNING

- Please carry out only the maintenance procedures described in this manual. If you require further assistance, please contact the nearest Yokogawa office.
- Care should be taken to prevent the build up of dust or other materials on the display glass and the name plate. To clean these surfaces, use a soft, dry cloth.

(e) Explosion Protected Type Instrument



WARNING

- Users of explosion proof instruments should refer first to section 2.9 (Installation of an Explosion Protected Instrument) of this manual.
- The use of this instrument is restricted to those who have received appropriate training in the device.
- Take care not to create sparks when accessing the instrument or peripheral devices in a hazardous location.

(f) Modification



WARNING

- Yokogawa will not be liable for malfunctions or damage resulting from any modification made to this instrument by the customer.

(g) Product Disposal

- The instrument should be disposed of in accordance with local and national legislation/regulations.

(h) Authorized Representative in EEA and the importer into the EU/EEA market

- In relation to the CE Marking, The authorised representative for this product in the EEA (European Economic Area) is:
Yokogawa Europe B.V.
Euroweg 2, 3825 HD Amersfoort, The Netherlands
and the importer for this product into the EU/EEA market via the YOKOGAWA sales channel is:
Yokogawa Europe B.V.
Euroweg 2, 3825 HD Amersfoort, The Netherlands

(i) Control of Pollution Caused by the Product

- Please refer to the IM 01C00A20-01ZH for information on our management status pursuant to “Control of Pollution caused by Electronic Information Products” of the People’s Republic of China. Please search for the IM number in the Yokogawa library
“ <https://www.yokogawa.com/library/> ”.

1.2 Warranty

- The warranty shall cover the period noted on the quotation presented to the purchaser at the time of purchase. Problems occurring during the warranty period shall basically be repaired free of charge.
 - If any problems are experienced with this instrument, the customer should contact the Yokogawa representative from which this instrument was purchased or the nearest Yokogawa office.
 - If a problem arises with this instrument, please inform us of the nature of the problem and the circumstances under which it developed, including the model specification and serial number. Any diagrams, data and other information you can include in your communication will also be helpful.
 - The party responsible for the cost of fixing the problem shall be determined by Yokogawa following an investigation conducted by Yokogawa.
- The purchaser shall bear the responsibility for repair costs, even during the warranty period, if the malfunction is due to:
 - Improper and/or inadequate maintenance by the purchaser.
 - Malfunction or damage due to a failure to handle, use, or store the instrument in accordance with the design specifications.
 - Use of the product in question in a location not conforming to the standards specified by Yokogawa, or due to improper maintenance of the installation location.
 - Failure or damage due to modification or repair by any party except Yokogawa or an approved representative of Yokogawa.
 - Malfunction or damage from improper relocation of the product in question after delivery.
 - Reason of force majeure such as fires, earthquakes, storms/floods, thunder/lightening, or other natural disasters, or disturbances, riots, warfare, or radioactive contamination.

2. Handling Cautions

This chapter provides important information on how to handle the transmitter. Read this carefully before using the transmitter.

The transmitters are thoroughly tested at the factory before shipment. When taking delivery of an instrument, visually check them to make sure that no damage occurred during shipment.

Also check that all transmitter mounting hardware shown in figure 2.1 is included. If the transmitter is ordered without the mounting bracket and the process connector, the transmitter mounting hardware will not be included. When the option code /BS is specified, a process connector for low pressure side will not be included. After checking the transmitter, carefully repack it in its box and keep it there until you are ready to install it.

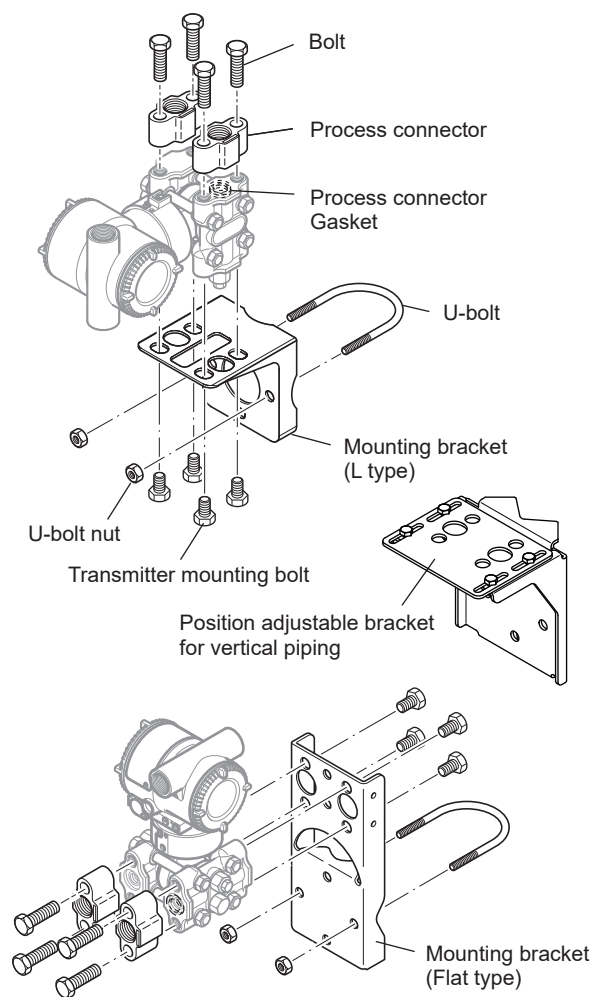


Figure 2.1 Transmitter Mounting Hardware

2.1 Model and Specifications Check

The model name and specifications are written on the name plate attached to the case.

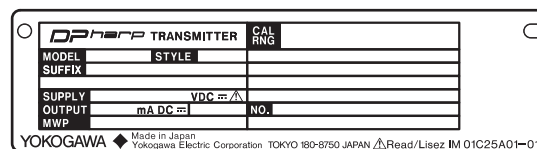


Figure 2.2 Name Plate (EJX110A)

2.2 Unpacking

Keep the transmitter in its original packaging to prevent it from being damaged during shipment. Do not unpack the transmitter until it reaches the installation site.

2.3 Storage

The following precautions must be observed when storing the instrument, especially for a long period.

- (a) Select a storage area which meets the following conditions:
 - It is not exposed to rain or subject to water seepage/leaks.
 - Vibration and shock are kept to a minimum.
 - It has an ambient temperature and relative humidity within the following ranges.

Ambient temperature:

–40* to 85°C without integral indicator

–30* to 80°C with integral indicator

* –15°C when /HE is specified.

Relative humidity:

0% to 100% R.H.

Preferred temperature and humidity:

approx. 25°C and 65% R.H.

- (b) When storing the transmitter, repack it carefully in the packaging that it was originally shipped with.
- (c) If the transmitter has been used, thoroughly clean the chambers inside the cover flanges, so that there is no process fluid remaining inside. Before placing it in storage, also make sure that the pressure-detector is securely connected to the transmitter section.

2.4 Selecting the Installation Location

The transmitter is designed to withstand severe environmental conditions. However, to ensure that it will provide years of stable and accurate performance, take the following precautions when selecting the installation location.

- (a) **Ambient Temperature**
Avoid locations subject to wide temperature variations or a significant temperature gradient. If the location is exposed to direct sunlight or radiant heat from plant equipment, provide adequate shade, thermal insulation and/or ventilation.
- (b) **Ambient Atmosphere**
Do not install the transmitter in a corrosive atmosphere. If this cannot be avoided, there must be adequate ventilation as well as measures to prevent the leaking of rain water and the presence of standing water in the conduits.
- (c) **Shock and Vibration**
Although the transmitter is designed to be relatively resistant to shock and vibration, an installation site should be selected where this is kept to a minimum.
- (d) **Installation of Explosion-protected Transmitters**
An explosion-protected transmitters is certified for installation in a hazardous area containing specific gas types. See subsection 2.9 "Installation of an Explosion-Protected Transmitters."

2.5 Pressure Connection



WARNING

- Never loosen the process connector bolts when an instrument is installed in a process. The device is under pressure, and a loss of seal can result in a sudden and uncontrolled release of process fluid.
- When draining toxic process fluids that have condensed inside the pressure detector, take appropriate steps to prevent the contact of such fluids with the skin or eyes and the inhalation of vapors from these fluids.

The following precautions must be observed in order to safely operate the transmitter under pressure.

- (a) Make sure that all the process connector bolts are tightened firmly.
- (b) Make sure that there are no leaks in the impulse piping.
- (c) Never apply a pressure higher than the specified maximum working pressure.

2.6 Waterproofing of Cable Conduit Connections

Apply a non-hardening sealant to the threads to waterproof the transmitter cable conduit connections. (See figure 6.8, 6.9 and 6.10.)

2.7 Restrictions on Use of Radio Transceivers



IMPORTANT

Although the transmitter has been designed to resist high frequency electrical noise, if a radio transceiver is used near the transmitter or its external wiring, the transmitter may be affected by high frequency noise pickup. To test this, start out from a distance of several meters and slowly approach the transmitter with the transceiver while observing the measurement loop for noise effects. Thereafter use the transceiver outside the range where the noise effects were first observed.

2.8 Insulation Resistance and Dielectric Strength Test

Since the transmitter has undergone insulation resistance and dielectric strength tests at the factory before shipment, normally these tests are not required. If the need arises to conduct these tests, heed the following:

- (a) Do not perform such tests more frequently than is absolutely necessary. Even test voltages that do not cause visible damage to the insulation may degrade the insulation and reduce safety margins.
- (b) Never apply a voltage exceeding 500 V DC (100 V DC with an internal lightning protector) for the insulation resistance test, nor a voltage exceeding 500 V AC (100 V AC with an internal lightning protector) for the dielectric strength test.
- (c) Before conducting these tests, disconnect all signal lines from the transmitter terminals. The procedure for conducting these tests is as follows:

• Insulation Resistance Test

- 1) Short-circuit the + and – SUPPLY terminals in the terminal box. In case of 1 to 5 V output, short-circuit the SUPPLY+, SUPPLY – and A (VOUT +) terminals.
- 2) Turn OFF the insulation tester. Then connect the insulation tester plus (+) lead wire to the shorted SUPPLY terminals and the minus (–) leadwire to the grounding terminal.
- 3) Turn ON the insulation tester power and measure the insulation resistance. The voltage should be applied as briefly as possible to verify that the insulation resistance is at least 20 MΩ.
- 4) After completing the test and being very careful not to touch exposed conductors disconnect the insulation tester and connect a 100 kΩ resistor between the grounding terminal and the short-circuiting SUPPLY terminals. Leave this resistor connected at least one second to discharge any static potential. Do not touch the terminals while it is discharging.

• Dielectric Strength Test

- 1) Short-circuit the + and – SUPPLY terminals in the terminal box. In case of 1 to 5 V output, short-circuit the SUPPLY+, SUPPLY – and A (VOUT +) terminals.
- 2) Turn OFF the dielectric strength tester. Then connect the tester between the shorted SUPPLY terminals and the grounding terminal. Be sure to connect the grounding lead of the dielectric strength tester to the ground terminal.
- 3) Set the current limit on the dielectric strength tester to 25 mA, then turn ON the power and gradually increase the test voltage from '0' to the specified voltage.
- 4) When the specified voltage is reached, hold it for one minute.
- 5) After completing this test, slowly decrease the voltage to avoid any voltage surges.

2.9 Installation of an Explosion-Protected Instrument

When products certified for explosion protection by FM, CSA, ATEX, or IECEx are used as explosion-protected instruments, installation, wiring, operating conditions, maintenance, and repair shall comply with the requirements of the applicable explosion-protection certifications and relevant laws and regulations.

Detailed requirements are described in the Explosion-Protected Instruments User's Manual IM 01C25A01-10EN, which is supplied with the product at shipment. When installing or using an explosion-protected instrument, be sure to refer to the applicable Explosion-Protected Instruments User's Manual and strictly follow its instructions. The information provided in this user's manual alone may not be sufficient to ensure compliance with explosion-protection certification requirements and relevant laws and regulations.

For products certified under other explosion-protection approvals, refer to the applicable explosion-protected user's manual supplied with the product.

The Explosion-Protected Instruments User's Manual IM 01C25A01-10EN is also available for download from the Yokogawa website library.

<https://www.yokogawa.com/library/>

2.10 EMC Conformity Standards

EN 61326-1 Class A, Table2

EN 61326-2-3

EN 61326-2-5 (for Fieldbus)

Immunity influence during the test

Differential pressure / Pressure: Output shift is specified within $\pm 1\%$ of 1/20 Max span.

Static pressure: Output shift is specified within $\pm 2\%$ of 1 MPa span.

Status Output Line*: Continues to operate without reversal.

*: Only for /AL option.

Condition for EMC test for FOUNDATION fieldbus type: The shield of the cable and the case are connected with a capacitor of 10 nF.



CAUTION

To meet EMC regulations, Yokogawa recommends that customers run signal wiring through metal conduits or use shielded twisted-pair cabling when installing EJX/EJA-E series transmitters in a plant.

This equipment is a Class A product, and it is designed for use in the industrial environment. Please use this instrument in the industrial environment only.

2.11 Pressure Equipment Directive (PED)

(1) General

- EJX/EJA-E Series pressure transmitters are categorized as piping under the pressure accessories section of directive 2014/68/EU, which corresponds to Article 4, Paragraph 3 of PED, denoted as Sound Engineering Practice (SEP).
- EJX110A-□MS, EJX110A-□HS, EJX110A-□VS, EJ□130□, EJ□440□, and EJA110E with /HG can be used above 200 bar and therefore considered as a part of a pressure retaining vessel where category III, Module H applies. These models with option code /PE3 conform to that category.

(2) Technical Data

- Models without /PE3
Article 4, Paragraph 3 of PED, denoted as Sound Engineering Practice (SEP).
- Models with /PE3
Module: H
Type of Equipment: Pressure Accessory-Vessel
Type of fluid: Liquid and Gas
Group of fluid: 1 and 2

Model	Capsule code	PS*1 (bar)	V (L)	PS.V (bar.L)	TS*2 (°C)	Category*3
EJA110E	M, H, V	160	0.01	1.6	-40 to 120	Article 4, Paragraph 3 (SEP)
EJ□110□	F, L					
EJX110A	M, H, V					
EJA110E with code /HG						
EJ□110□ with code /PE3	M, H, V	250	0.01	2.5	-29 to 120	III
EJ□130□	M, H, V	500	0.01	5.0	-40 to 120	Article 4, Paragraph 3 (SEP)
EJ□130□ with code /PE3	M, H, V	500	0.01	5.0		III
EJ□310□	L	160	0.01	1.6	-40 to 100	Article 4, Paragraph 3 (SEP)
	M, A, B				-40 to 120	
EJ□430□	H, A, B	160	0.01	1.6	-40 to 120	Article 4, Paragraph 3 (SEP)
EJ□440□	C, D	500	0.01	5.0	-40 to 120	Article 4, Paragraph 3 (SEP)
EJ□440□ with code /PE3	C, D	500	0.01	5.0		III

*1: PS is maximum pressure for vessel itself based on Pressure Equipment Directive 2014/68/EU. Refer to General Specification for maximum working pressure of a transmitter.

*2: TS is process temperature limits for vessel itself based on Pressure Equipment Directive 2014/68/EU. Refer to General Specification for temperature limits of a transmitter.

*3: Referred to Table 1 covered by ANNEX II of EC Directive on Pressure Equipment Directive 2014/68/EU.

(3) Operation**CAUTION**

- The temperature and pressure of fluid should be maintained at levels that are consistent with normal operating conditions.
- The ambient temperature should be maintained at a level that is consistent with normal operating conditions.
- Please take care to prevent water hammer and the like from inducing excessive pressures in pipes and valves. If phenomena are likely, install a safety valve or take some other appropriate measure to prevent pressure from exceeding PS.
- Take appropriate measures at the device or system level to protect transmitters if they are to be operated near an external heat source.

2.12 EU RoHS Directive

Applicable standard: EN IEC 63000

Applicable production sites are shown below.

The condition of the RoHS compliant production sites are as follows:

Japan, Germany, Saudi Arabia,
UAE, India, USA

The production sites can be confirmed by the serial number shown in the frame of "NO." in the name plate of the product.

Serial numbers (9 letters): **AA**nnnnnnnn

AA: Identification code of production site

Japan: Use "91" or "90"

USA: Use "U1"

Germany: Use "D1"

Saudi Arabia: Use "Y3"

UAE: Use "Y4"

India: Use "Y1"

2.13 Safety Requirement Standards

Applicable standard: EN 61010-1, C22.2 No.61010-1

(1) Pollution Degree 2

"Pollution degree" describes the degree to which a solid, liquid, or gas which deteriorates dielectric strength or surface resistivity is adhering. "2" applies to normal indoor atmosphere. Normally, only non-conductive pollution occurs. Occasionally, however, temporary conductivity caused by condensation must be expected.

(2) Installation Category I

"Overvoltage category (Installation category)" describes a number which defines a transient overvoltage condition. It implies the regulation for impulse withstand voltage. "I" applies to electrical equipment which is supplied from the circuit when appropriate transient overvoltage control means (interfaces) are provided.

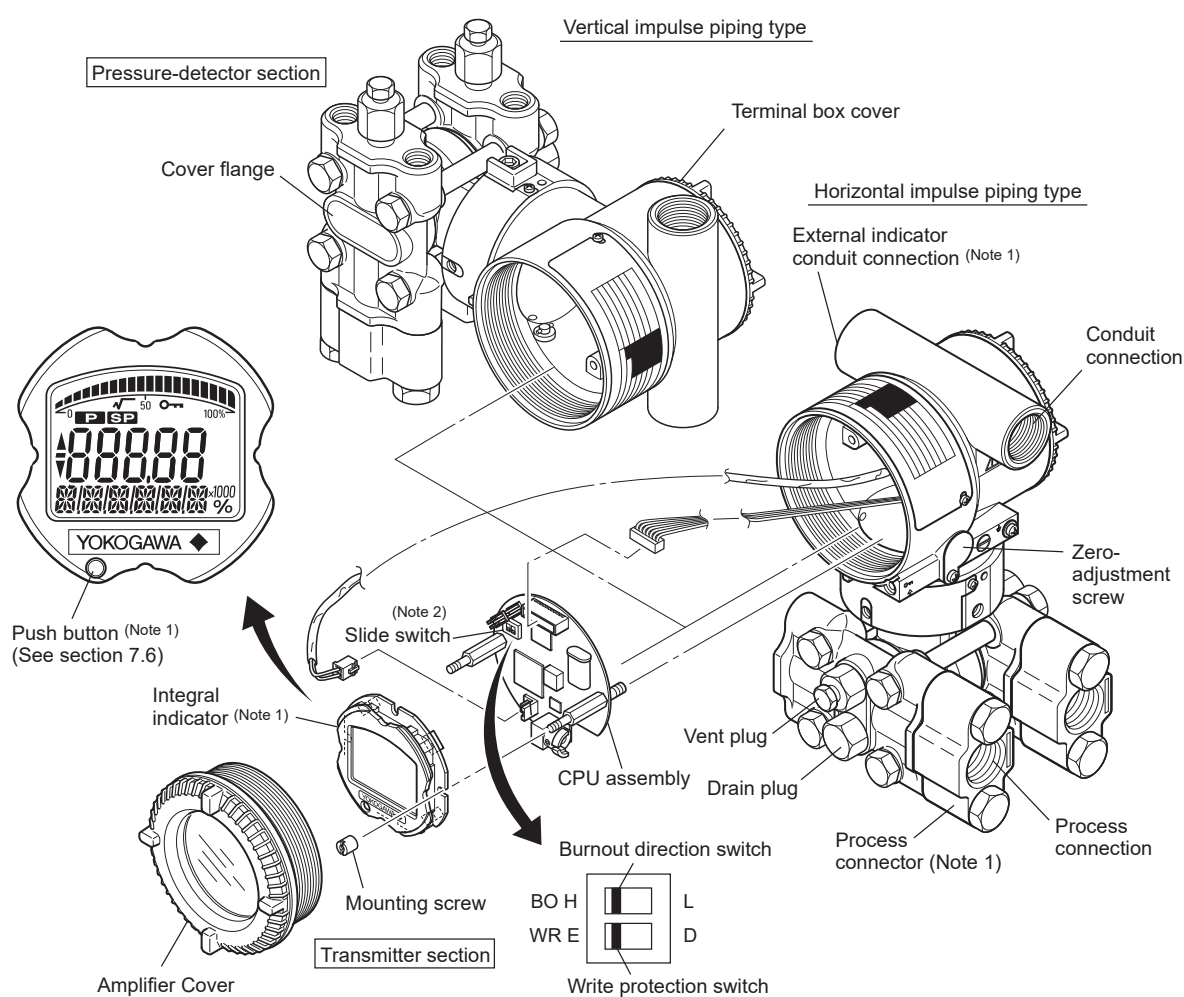
(3) Indoor/Outdoor use

It can be used in indoors and outdoors.

(4) Altitude of installation site

Max. 5,000 m above sea level

3. Component Names



Burnout direction switch (BO)			Hardware write protection switch (WR)		
Burnout Direction Switch Position	H  L	H  L	Write Protection Switch Position	H  L	H  L
Burnout Direction	HIGH	LOW	Write Protection	NO (Write enabled)	YES (Write disabled)

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- Note 1: See General Specifications sheet of each product for details. A process connector will not be applied for lower side of EJ□310□, EJ□430□, and EJ□440□. A process connector will not be applied for lower side of EJ□110□ and EJ□120□ when option code /BS is specified.
- Note 2: Applicable for BRAIN/HART communication type. Set the switches as shown in the figure above to set the burn-out direction and write protection. The Burnout switch is set to the H side for delivery (unless option code /C1 or /C2 is specified in the order), and the hardware write protection switch is set to E side. The setting of the switches can be confirmed via communication. An external zero adjustment screw can only be disabled by communication. To disable the screw, set a parameter before activating the hardware write protect function. The zero adjustment screw is disabled upon shipment when option code /CJ or /CK is specified. See each communication manual.

Figure 3.1 Component Names

Table 3.1 Display Symbol

Display Symbol	Meaning of Display Symbol
	Display mode is 'square root'. (Display is not lit when 'linear' mode.)
	The output signal being zero-adjusted is increasing. Besides, this symbol lights when local parameter setting is in progress.
	The output signal being zero-adjusted is decreasing. Besides, this symbol lights when local parameter setting is in progress.
	Write protect function is enabled.

F0302.ai

4. Installation

4.1 Precautions

Before installing the transmitter, read the cautionary notes in section 2.4, "Selecting the Installation Location." For additional informations and limitations according to the selected model, suffix and option codes, please find information in each General Specifications sheet.



IMPORTANT

- When welding piping during construction, take care not to allow welding currents to flow through the transmitter.
- Do not step on this instrument after installation.
- For the EJ□430□ and EJ□440□, the atmospheric opening is located on the low pressure side cover flange. The opening must not face upward.
- When installing a process connector, make sure that no foreign matter has adhered to the seal surface of the gasket (or O-ring) before assembly. If foreign matter adheres, it may lead to leaks. In the case of O-rings, it is known that the sealing performance can be improved by applying grease. If grease is not used, make sure that no foreign matter is attached.

4.2 Mounting

- The transmitter is shipped with the process connection, according to the ordering specifications. To change the orientation of the process connections, refer to section 4.3.
- With differential pressure transmitters, the distance between the impulse piping connection ports is usually 54 mm (Figure 4.1). By changing the orientation of the process connector, the dimension can be changed to 51 mm or 57 mm.
- The transmitter can be mounted on a nominal 2-inch pipe using the mounting bracket supplied, as shown in figure 4.2 and 4.3. The transmitter can be mounted on either a horizontal or a vertical pipe.
- When mounting the bracket on the transmitter, tighten the (four) bolts that hold the transmitter with a torque of approximately 39 N·m {4kgf·m}.

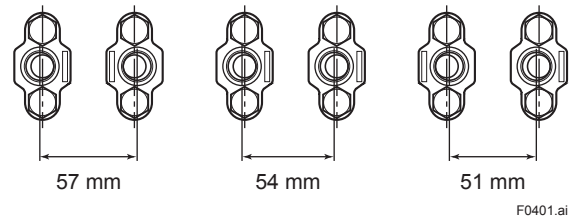
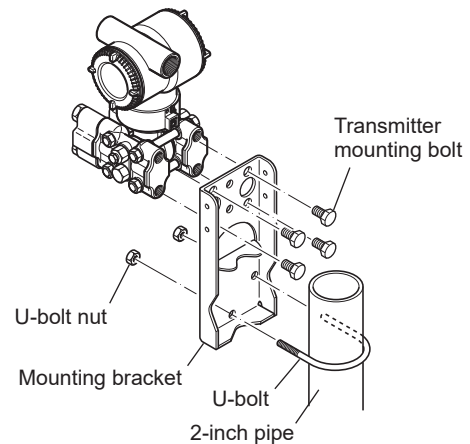


Figure 4.1 Process Connector Impulse Piping Connection Distances for Differential Pressure Transmitters

Figure 4.1 and 4.2 shows the mounting of the transmitter for horizontal piping and vertical piping with using the mounting bracket. The transmitters with the installation code -U (Universal flange) can be used for either type of mounting.

Vertical pipe mounting



Horizontal pipe mounting

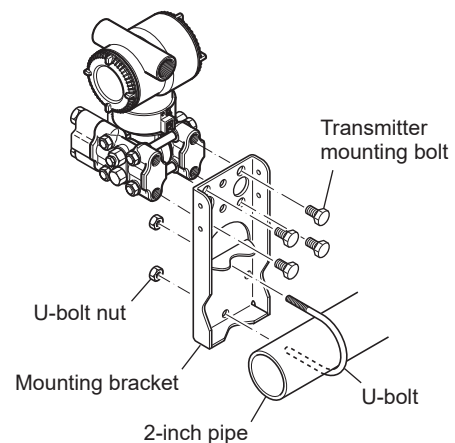
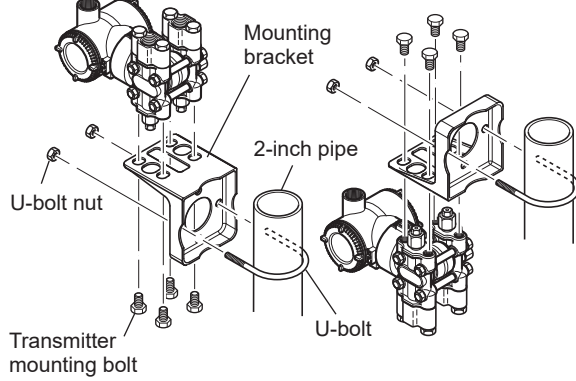


Figure 4.2 Transmitter Mounting (Horizontal Impulse Piping Type)

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Vertical pipe mounting
(Process connector upside)

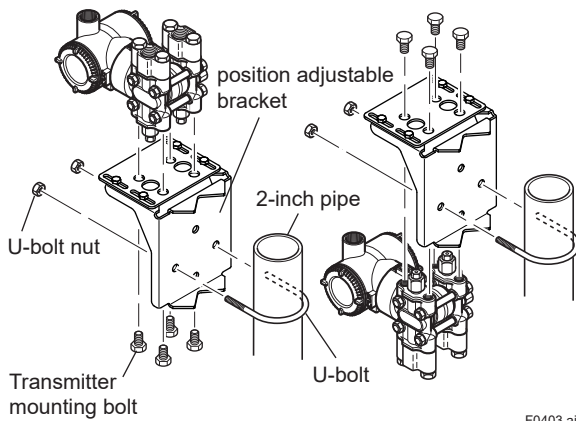
Vertical pipe mounting
(Process connector downside)



■ In the case of position adjustable bracket

Vertical pipe mounting
(Process connector upside)

Vertical pipe mounting
(Process connector downside)



F0403.ai

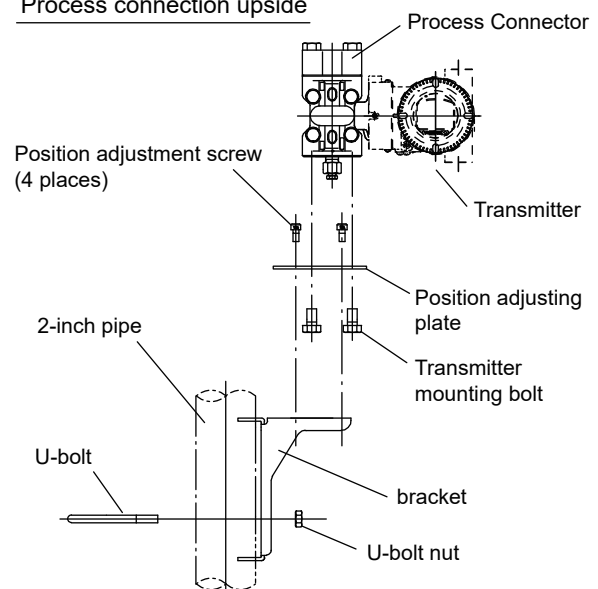
Figure 4.3 Transmitter Mounting (Vertical Impulse Piping Type)

4.2.1 How to install the position adjustable bracket

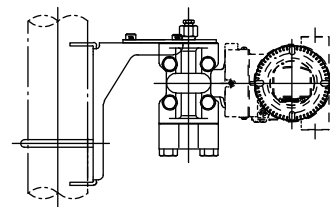
The procedures for mounting the transmitter with the position adjustable bracket (L type) are described below. (See figure 4.4)

- (1) Securely fix the transmitter and the position adjustable plate using the transmitter mounting bolts. Tighten the transmitter mounting four bolts with a torque of approximately 39 N·m {4 kgf·m}.
- (2) Attach the bracket to the 2-inch pipe while adjusting the height to the process piping.
- (3) Loosen the position adjustment screw and slide the position adjustable plate on the bracket to align the position of the 2-inch pipe and the process piping.* 1
- (4) Tighten each of the position adjustment four bolts to firmly fix the position adjustable plate and the bracket. After fixing, check that it does not move.

Process connection upside



Process connection downside

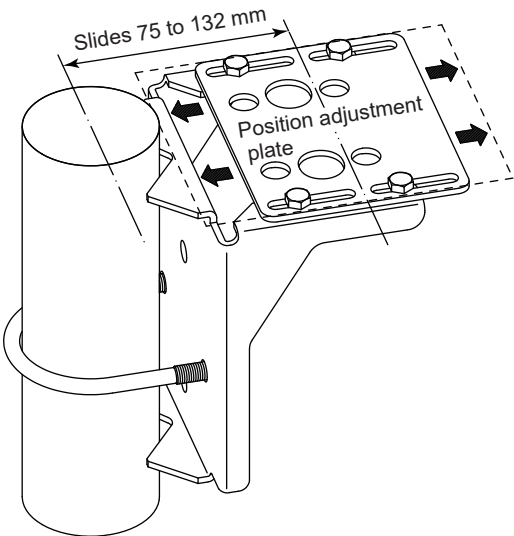


Mount the bracket in the direction as described in the figure.

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Figure 4.4 Mounting with the position adjustable bracket

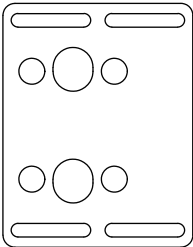
- *1: The position adjustment plate can be adjusted in the range of 75 to 132 mm depending on the mounting direction to the bracket. The adjustment range varies depending on the model and specifications. (See figures 4.5 to 4.6)
- *2: When the EJX130A is specified as “process connector downside”, the position adjustment plate can be used only in the inverted state shown in figure 4.6.



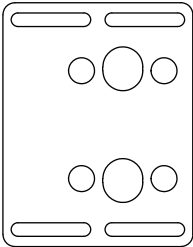
F0410.ai

Figure 4.5 Adjustable range of the plate

	The state of the plate seen from the above	The state of the plate seen from the side (EJX110A)
Plate in minimum position (75 mm)		
Plate in maximum position (110 mm)		
Inverted plate in maximum position (132 mm)		



Inverted



F0411.ai

Figure 4.6 The adjustable range by the mounting direction of the plate

4.3 Changing the Process Connection

The transmitter is shipped with the process connection specified at the time of ordering. To change the process connection, the drain (vent) plug must be repositioned.

To reposition a drain (vent) plug, use a wrench to slowly and gently unscrew it. Then, remove and remount it on the opposite side. Wrap sealing tape around the drain (vent) plug threads (*1 in the figure below), and apply a lubricant to the threads of the drain (vent) screw(s) (*2 below). To tighten the drain (vent) plugs, apply a torque of 34 to 39 N·m (3.5 to 4 kgf·m). Process connector bolts are to be tightened uniformly to a torque shown in table 4.1.

Table 4.1 Torque

Model	EJ□110□ EJ□120□ EJ□130□ EJ□310□ EJ□430□	EJ□440□	
		C capsule	D capsule
Torque(N·m) {kgf·m}	39 to 49 {4 to 5}	49 to 59 {5 to 6}	

Vertical impulse piping type

Horizontal impulse piping type

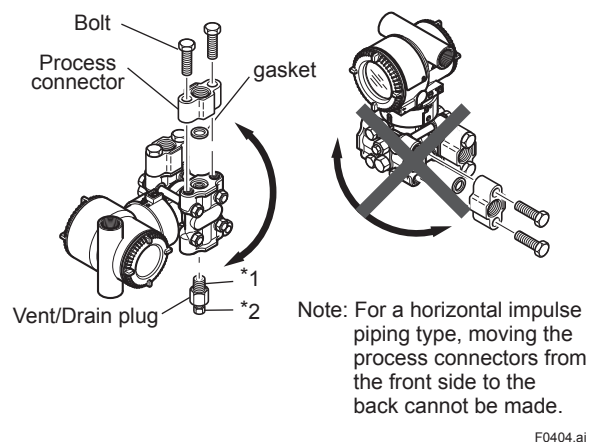


Figure 4.7 Changing Process Connection



IMPORTANT

When installing a process connector, make sure that no foreign matter has adhered to the seal surface of the gasket (or O-ring) before assembly. If foreign matter adheres, it may lead to leaks. In the case of O-rings, it is known that the sealing performance can be improved by applying grease. If grease is not used, make sure that no foreign matter is attached.

4.4 Swapping the High/Low-pressure Side Connection



IMPORTANT

This section is applicable only for EJ□110□, EJ□120□, and EJ□130□ differential transmitters, and not applicable for gauge or absolute pressure transmitters.

4.4.1 Rotating Pressure-detector Section 180°

This procedure can be applied only to a transmitter with a vertical impulse piping type.

The procedure below can be used to turn the pressure detector assembly 180°. Perform this operation in a maintenance shop with the necessary tools laid out and ready for use, and then install the transmitter in the field after making the change.

- 1) Use an Allen wrench (JIS B4648, nominal 2.5 mm) to remove the two setscrews at the joint between the pressure-detector section and transmitter section.
- 2) Leaving the transmitter section in position, rotate the pressure-detector section 180°.
- 3) Tighten the two setscrews to fix the pressure-detector section and transmitter section together (at a torque of 1.5 N·m).

Reposition the process connector and drain (vent) plugs to the opposite side as described in subsection 4.3.

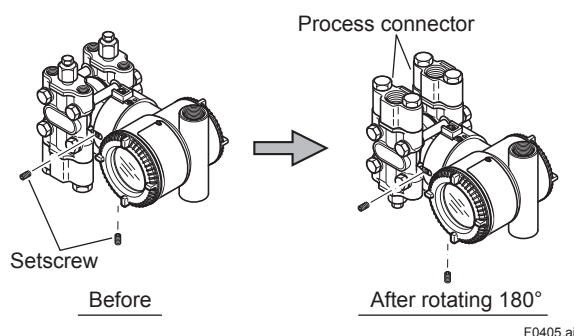


Figure 4.8 Before and After Modification

4.4.2 Using the Communicator

This method is applicable only to the Model EJ□110□, EJ□120□, and EJ□130□.

With a communicator, you can change which process connection is used as the high-pressure side without mechanically rotating the pressure-detector section 180° as described in subsection 4.4.1. To change, call parameter 'D15: H/L SWAP' for BRAIN Communication or 'H/L swap' for HART Communication and select REVERSE (right side: low pressure; left side: high pressure) or select NORMAL to change back to normal (right side: high pressure; left side: low pressure).

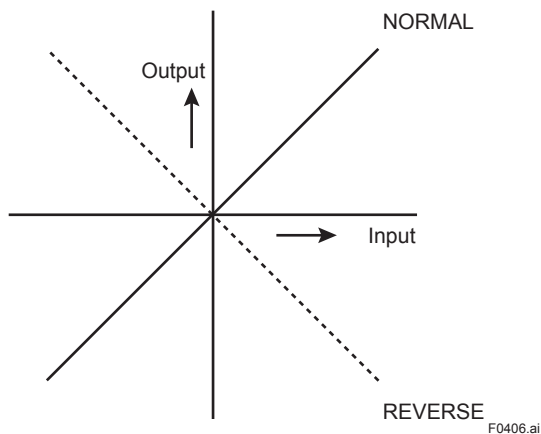


Figure 4.9 Input/Output Relationship



IMPORTANT

Since the H/L label plate on the capsule assembly will remain unchanged, use this function only when you cannot switch the impulse piping. If the 'H/L SWAP' parameter setting is changed, the input/output relationship is reversed as shown in figure 4.9; be sure this is understood by all.

4.5 Rotating Transmitter Section

The transmitter section can be rotated approximately 360° and can be fixed at any angle within the above range. (The direction of the rotation is depending on the configuration of the instrument.) Note that there is a stopper which prevents the transmitter section from being rotated more than 360°.

- 1) Remove the two setscrews that fasten the transmitter section and capsule assembly, using the Allen wrench.
- 2) Rotate the transmitter section slowly and stop it at designated position.
- 3) Tighten the two setscrews to a torque of 1.5 N·m.



WARNING

In the case of the explosion-proof/flameproof type transmitter, do not rotate the transmitter part in the hazardous area while the transmitter is energized.

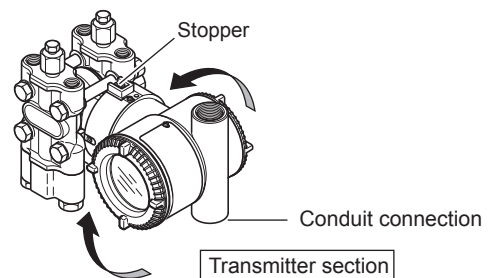


IMPORTANT

Do not rotate the transmitter section more than the above limit.

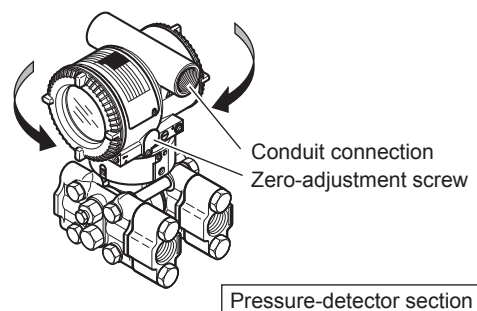
Vertical impulse piping type

Pressure-detector section



Horizontal impulse piping type

Transmitter section



F0407.ai

Figure 4.10 Rotating Transmitter Section (Left Side High Pressure Type)

4.6 Changing the Direction of Integral Indicator



IMPORTANT

Always turn OFF power, release pressure and remove a transmitter to non-hazardous area before disassembling and reassembling an indicator.

An integral indicator can be installed in the following three directions. Follow the instructions in section 8.4 for removing and attaching the integral indicator.

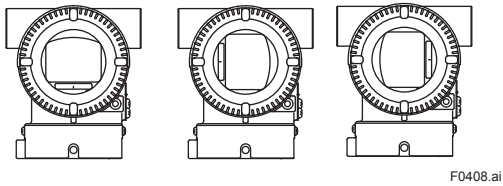


Figure 4.11 Integral Indicator Direction

5. Installing Impulse Piping

5.1 Impulse Piping Installation Precautions

The impulse piping that connects the process outputs to the transmitter must convey the process pressure accurately. If, for example, gas collects in a liquid-filled impulse line, or the drain for a gas-filled impulse line becomes plugged, it will not convey the pressure accurately. Since this will cause errors in the measurement output, select the proper piping method for the process fluid (gas, liquid, or steam). Pay careful attention to the following points when routing the impulse piping and connecting the impulse piping to a transmitter.

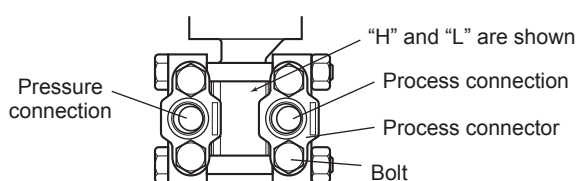
5.1.1 Connecting Impulse Piping to a Transmitter

(1) Check the High and Low Pressure Connections on the Transmitter (Figure 5.1)

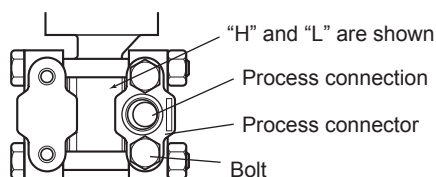
Symbols “H” and “L” have been placed on the capsule assembly to indicate high and low pressure side. With differential pressure transmitters, connect the high pressure side impulse line to the “H” side, and the low pressure side impulse line to the “L” side. When option code /BS is specified, piping or process connector can't be connected to the low pressure side.

With gauge/absolute pressure transmitters, connect the impulse line to the ‘H’ side.

Differential Pressure Transmitter



Gauge/Absolute Pressure Transmitters



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Figure 5.1 “H” and “L” Symbols on a Capsule Assembly

(2) Changing the Process Connector Piping Connections (Figure 4.1) (for differential pressure transmitters)

The impulse piping connection distances can be changed between 51 mm, 54 mm and 57 mm by changing the orientation of the process connectors. This is convenient for aligning an impulse line with a process connectors.

(3) Tightening the Process Connector Mounting Bolts

After connecting an impulse line, tighten the process connector mounting bolts uniformly.

(4) Removing the Impulse Piping Connecting Port Dustproof Cap

The impulse piping connecting port on the transmitter may be covered with a plastic cap to keep out dust. This cap must be removed before connecting the line. (Be careful not to damage the threads when removing this cap. Never insert a screwdriver or other tool between the cap and port threads to remove the cap.)

(5) Connecting the Transmitter and 3-Valve Manifold (for differential pressure transmitters)

A 3-valve manifold consists of two stop valves to block process pressure and an equalizing valve to equalize the pressures on the high and low pressure sides of the transmitter. Such a manifold makes it easier to disconnect the transmitter from the impulse piping, and is convenient when adjusting the transmitter zero point.

There are two 3-valve manifold types: the pipe-mounting type and the direct-mounting type; care should be taken with respect to the following points when connecting the manifold to the transmitter.

■ Pipe-Mounting Type 3-Valve Manifold (Figure 5.2)

- 1) Screw nipples into the connection ports on the transmitter side of the 3-valve manifold, and into the impulse piping connecting ports on the process connectors. (To maintain proper sealing, wind sealing tape around the nipple threads.)
- 2) Mount the 3-valve manifold on the 2-inch pipe by fastening a U-bolt to its mounting bracket. Tighten the U-bolt nuts only lightly at this time.
- 3) Install the pipe assemblies between the 3-valve manifold and the process connectors and lightly tighten the ball head lock nuts. (The ball-shaped ends of the pipes must be handled carefully, since they will not seal properly if the ball surface is scratched or otherwise damaged.)
- 4) Now tighten the nuts and bolts securely in the following sequence:

Process connector bolts → transmitter-end ball head lock nuts → 3-valve manifold ball head lock nuts → 3-valve manifold mounting bracket U-bolt nuts

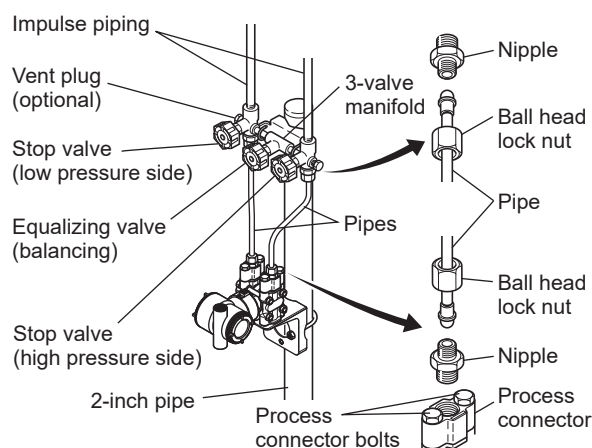


Figure 5.2 3-Valve Manifold (Pipe-Mounting Type)

■ Direct-Mounting Type 3-Valve Manifold (Figure 5.3)

- 1) Mount the 3-valve manifold on the transmitter. (When mounting, use the two gaskets and the four bolts provided with the 3-valve manifold. Tighten the bolts evenly.)
- 2) Mount the process connectors and gaskets on the top of the 3-valve manifold (the side on which the impulse piping will be connected).

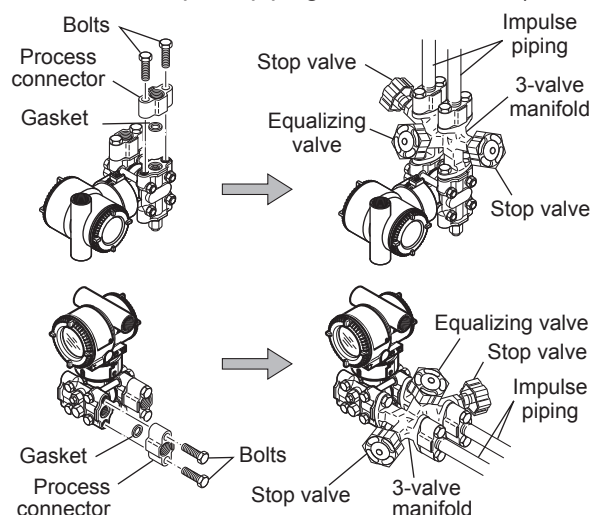


Figure 5.3 3-Valve Manifold (Direct-Mounting Type)



NOTE

After completing the connection of the transmitter and 3-valve manifold, be sure to CLOSE the low pressure and high pressure stop valves, OPEN the equalizing valve, and leave the manifold with the equalizing valve OPEN.

You must do this in order to avoid overloading the transmitter from either the high or the low pressure side when beginning operation.

This instruction must also be followed as part of the startup procedure (chapter 7.)

5.1.2 Routing the Impulse Piping

(1) Process Pressure Tap Angles

If condensate, gas, sediment or other extraneous material in the process piping gets into the impulse piping, pressure measurement errors may result. To prevent such problems, the process pressure taps must be angled as shown in figure 5.4 according to the kind of fluid being measured.



NOTE

- If the process fluid is a gas, the taps must be vertical or within 45° either side of vertical.
- If the process fluid is a liquid, the taps must be horizontal or below horizontal, but not more than 45° below horizontal.
- If the process fluid is steam or other condensing vapor, the taps must be horizontal or above horizontal, but not more than 45° above horizontal.

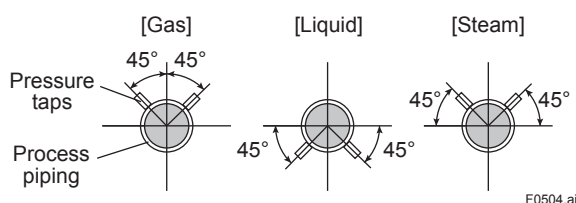


Figure 5.4 Process Pressure Tap Angle (For Horizontal Piping)

(2) Position of Process Pressure Taps and Transmitter

If condensate (or gas) accumulates in the impulse piping, it should be removed periodically by opening the drain (or vent) plugs. However, this will generate a transient disturbance in the pressure measurement, and therefore it is necessary to position the taps and route the impulse piping so that any extraneous liquid or gas generated in the leadlines returns naturally to the process piping.

- If the process fluid is a gas, then as a rule the transmitter must be located higher than the process pressure taps.
- If the process fluid is a liquid or steam, then as a rule the transmitter must be located lower than the process pressure taps.

(3) Impulse Piping Slope

The impulse piping must be routed with only an upward or downward slope. Even for horizontal routing, the impulse piping should have a slope of at least 1/10 to prevent condensate (or gases) from accumulating in the pipes.

(4) Temperature Difference Between Impulse Lines (for differential pressure transmitters)

If there is a temperature difference between the high and low impulse lines, the density difference of the fluids in the two lines will cause an error in the measurement pressure. When measuring flow, impulse lines must be routed together so that there is no temperature difference between them.

(5) Condensate Pots for Steam Flow Measurement (for differential pressure transmitters)

If the liquid in the impulse piping repeatedly condenses or vaporizes as a result of changes in the ambient or process temperature, this will cause a difference in the fluid head between the high pressure and low pressure sides. To prevent measurement errors due to these head differences, condensate pots are used when measuring steam flow.

(6) Preventing Wind Speed Effects in Very Low Differential Pressure Measurement (for differential pressure transmitters)



IMPORTANT

When using a differential pressure transmitter to measure very low pressures (draft pressure), the low pressure connection port is left open to atmospheric pressure (the reference pressure). Any wind around the differential pressure transmitter will therefore cause errors in the measurement. To prevent this, it will be necessary either to enclose the transmitter in a box, or to connect an impulse line to the low pressure side and insert its end into a wind-excluding pot (cylindrical with a base plate).

(7) Preventing Freezing

If there is any risk that the process fluid in the impulse piping or transmitter could freeze, use a steam jacket or heater to maintain the temperature of the fluid.



NOTE

After completing the connections, close the valves on the process pressure taps (main valves), the valves at the transmitter (stop valves), and the impulse piping drain valves, so that condensate, sediment, dust and other extraneous material cannot enter the impulse piping.

5.2 Impulse Piping Connection Examples

Figure 5.5 and 5.6 show examples of typical impulse piping connections. Before connecting the transmitter to the process, study the transmitter installation location, the process piping layout, and the characteristics of the process fluid (corrosiveness, toxicity, flammability, etc.), in order to make appropriate changes and additions to the connection configurations.

Note the following points when referring to these piping examples.

- The high pressure connecting port on the transmitter is shown on the right (as viewed from the front).
- The transmitter impulse piping connection is shown for a vertical impulse piping connection configuration in which the direction of connection is either upwards or downwards.
- If the impulse line is long, bracing or supports should be provided to prevent vibration.
- The impulse piping material used must be compatible with the process pressure, temperature, and other conditions.
- A variety of process pressure tap valves (main valves) are available according to the type of connection (flanged, screwed, welded), construction (globe, gate, or ball valve), temperature and pressure. Select the type of valve most appropriate for the application.

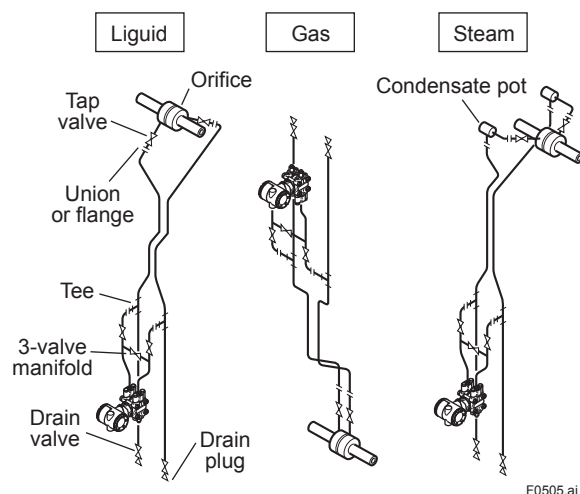


Figure 5.5 Impulse Piping Connection Examples for Differential Pressure Transmitters

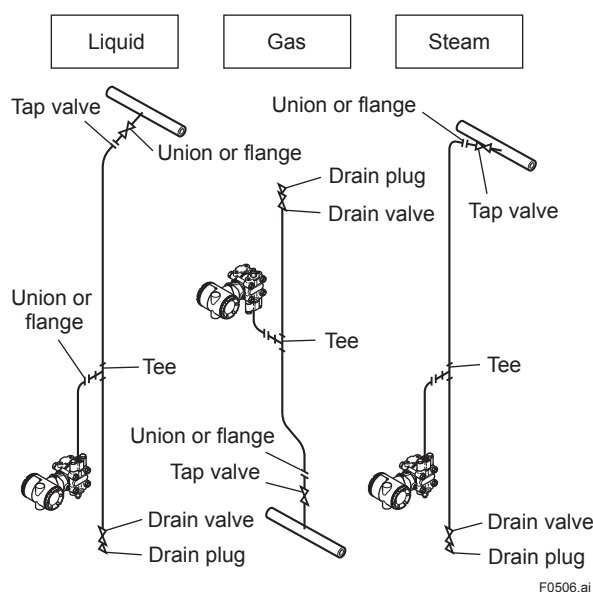


Figure 5.6 Impulse Piping Connection Examples for Gauge/absolute Pressure Transmitters

6. Wiring

6.1 Wiring Precautions



IMPORTANT

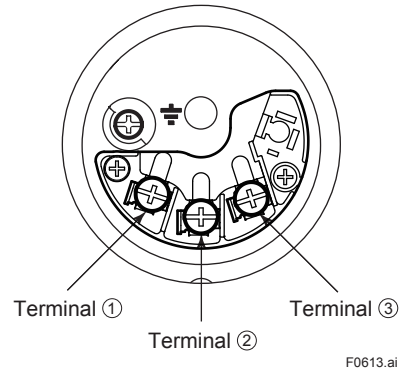
- Lay wiring as far as possible from electrical noise sources such as large capacity transformers, motors, and power supplies.
- Remove the electrical connection dust cap (plug) before wiring.
- All threaded parts must be treated with waterproofing sealant. (A non-hardening sealant is recommended.)
- To prevent noise pickup, do not pass signal and power cables through the same ducts.
- Explosion-protected instruments must be wired in accordance with specific requirements (and, in certain countries, legal regulations) in order to preserve the effectiveness of their explosion-protected features.
- The terminal box cover is locked by an Allen head bolt (a shrouding bolt) on ATEX flameproof type transmitters. When the shrouding bolt is driven clockwise using an Allen wrench, it goes in. The cover lock can then be released and the cover can be opened by hand. See subsection 8.4 “Disassembly and Reassembly” for details.
- Plug and seal an unused conduit connection.

6.2 Selecting the Wiring Materials

- Use stranded leadwires or cables which are the same as or better than 600 V grade PVC insulated wire (JIS C3307) or its equivalent.
- Use shielded wires in areas that are susceptible to electrical noise.
- In areas with higher or lower ambient temperatures, use appropriate wires or cables.
- In environment where oils, solvents, corrosive gases or liquids may be present, use wires or cables that are resistant to such substances.
- It is recommended that crimp-on solderless terminal lugs (for 4 mm screws) with insulating sleeves be used for leadwire ends.

6.3 Connections of External Wiring to Terminal Box

• Terminal Configuration



• Terminal Wiring for 4 to 20 mA output, FOUNDATION Fieldbus type, and PROFIBUS PA type.

SUPPLY	+	①	Power supply and output terminals
	-	②	
CHECK	+	③	External indicator (ammeter) terminals*1*2
or	-	②	
ALARM	+	③	Status contact output terminals*2 (when /AL is specified)
	-	②	
			Ground terminal

*1: When using an external indicator or check meter, the internal resistance must be 10 Ω or less. A check meter or indicator cannot be connected when /AL option is specified.

*2: Not available for FOUNDATION Fieldbus and PROFIBUS PA communication types.

• Terminal Wiring for 1 to 5 V output

SUPPLY	+	①	Power supply terminals
	-	②	
VOUT	+	③	1 to 5 V DC with HART communication terminals
	-	②	
			Ground terminal

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Figure 6.1 Terminal

6.3.1 Power Supply Wiring Connection



IMPORTANT

Connecting with the commercial AC power supply will damage the device. Be sure to use the DC power supply in the predetermined range.

Connect the power supply wiring to the SUPPLY + and – terminals. When /AL is specified, also refer to subsection 6.3.5.

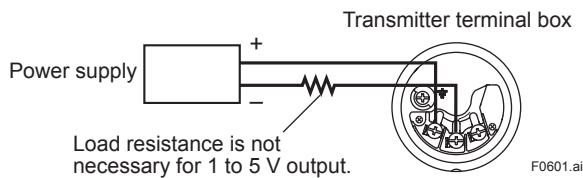


Figure 6.2 Power Supply Wiring Connection

6.3.2 External Indicator Connection

Available only for 4 to 20mA output type and when /AL is not specified.

Connect wiring for external indicators to the CHECK (+) and SUPPLY – terminals.

(Note) Use a external indicator whose internal resistance is 10 Ω or less.

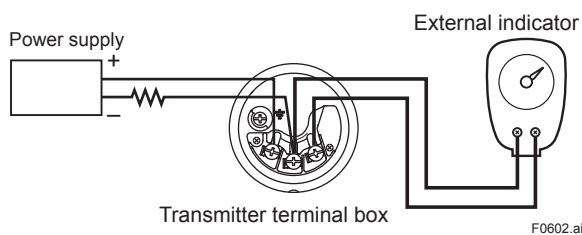


Figure 6.3 External Indicator Connection

6.3.3 Communicator Connection

- 4 to 20 mA output, BRAIN / HART



IMPORTANT

Analog output may change temporally in connecting with BRAIN terminal due to an initial current flowed to it. To prevent communication signal affecting the upper system, it is recommended to install a low-pass filter (approximately 0.1s)

Connect the BT200 or HART Communicator to the SUPPLY + and – terminals. (Use hooks.)

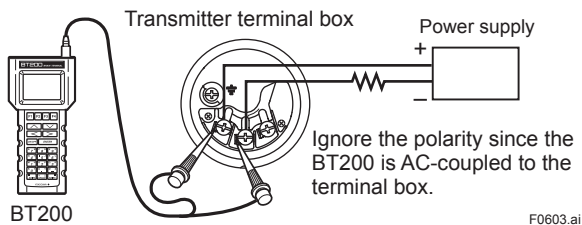


Figure 6.4 BT200 Connection

- 1 to 5 V output, HART
Connect the HART communicator or configuration tool to the SUPPLY - and VOUT (+) terminals. (Use hooks.)

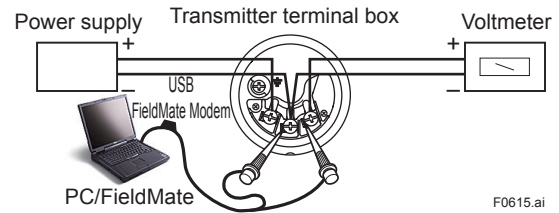


Figure 6.5 Four wire connection

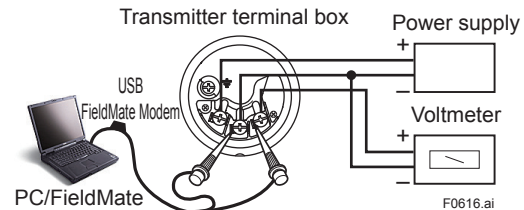


Figure 6.6 Three wire connection

6.3.4 Check Meter Connection

Available only for 4 to 20mA output type and when /AL is not specified.

Connect the check meter to the CHECK (+) and SUPPLY – terminals. (Use hooks.)

- A 4 to 20 mA DC output signal from the CHECK (+) and SUPPLY – terminals.

(Note) Use a check meter whose internal resistance is 10 Ω or less.

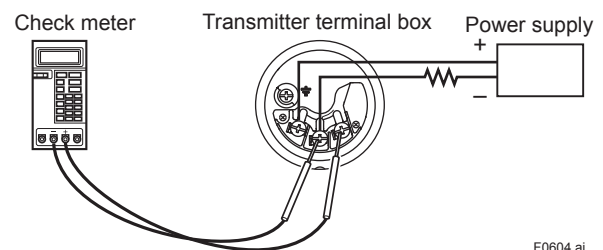


Figure 6.7 Check Meter Connection

6.3.5 Status Output Connection

When option code /AL is specified, connect the external wiring as shown in Figure 6.5.

To configure and activate the process alarm function and status output, it is necessary to set some parameters. Refer to each communication manual for procedures.

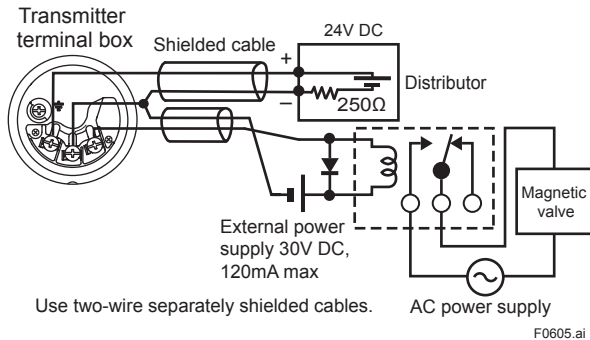


Figure 6.8 Status Output Connection

6.4 Wiring

6.4.1 Loop Configuration

Since the DPharp uses a two-wire transmission system for 4 to 20 mA output, signal wiring is also used as power wiring.

DC power is required for the transmitter loop. The transmitter and distributor are connected as shown below.

For details of the power supply voltage and load resistance, see section 6.6.

For 1 to 5 V output, three or four wire system is used. See (3).

(1) 4 to 20 mA output, General-use Type and Flameproof Type

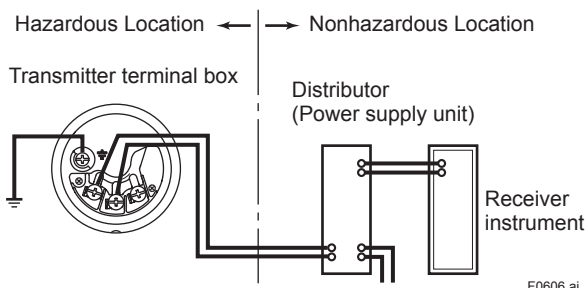


Figure 6.9 Connection between Transmitter and Distributor

(2) 4 to 20 mA output, Intrinsically Safe Type

With the intrinsically safe type, a safety barrier must be included in the loop.

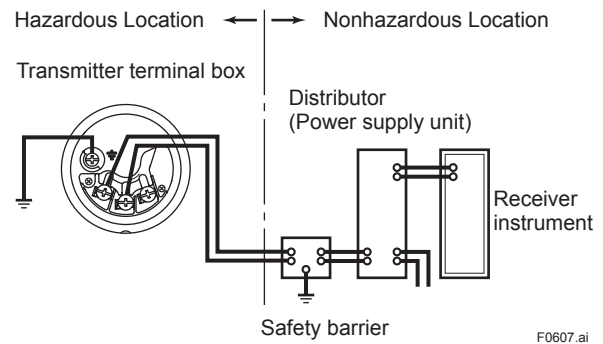


Figure 6.10 Connection between Transmitter and Distributor

(3) 1 to 5 V output

Either three or four wire system is used.

Power supply line and 1 to 5 V signal line commonly use the SUPPLY - terminal.



NOTE

With three wire connection, the cable length may affect the measurement accuracy of the output signal.

In either three-wire or four-wire connection, recommended wiring distance is 200 m or less, and the use of shielded cable is recommended.

■ Three wire connection

For three wire system, a negative wiring shall be commonly used for power supply and signal line.

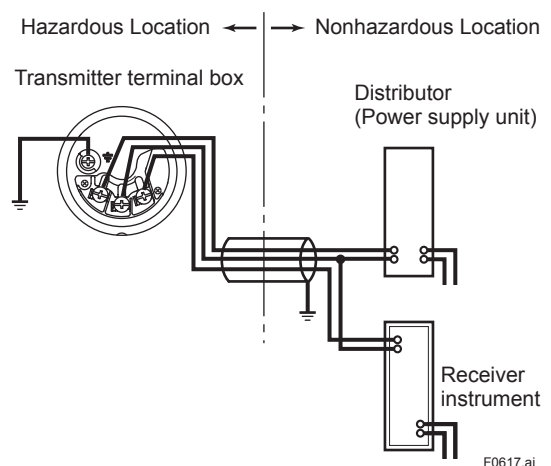


Figure 6.11 Connection between Transmitter, Distributor and Receiver

- Four wire connection
Fasten the negative side wiring of both power supply and signal line to the SUPPLY - terminal.

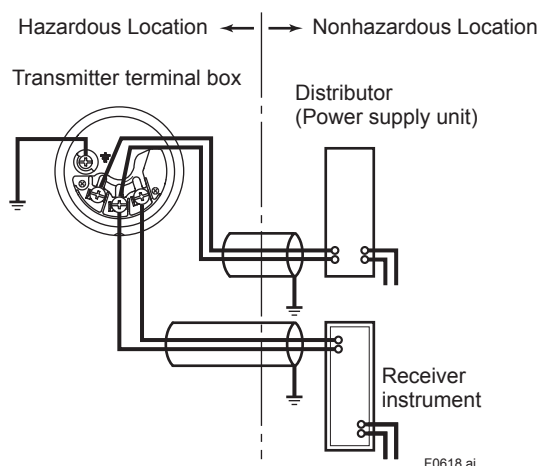


Figure 6.12 Connection between Transmitter, Distributer and Receiver

6.4.2 Wiring Installation

(1) General-use Type and Intrinsically Safe Type

With the cable wiring, use a metallic conduit or waterproof glands.

- Apply a non-hardening sealant to the terminal box connection port and to the threads on the flexible metal conduit for waterproofing.

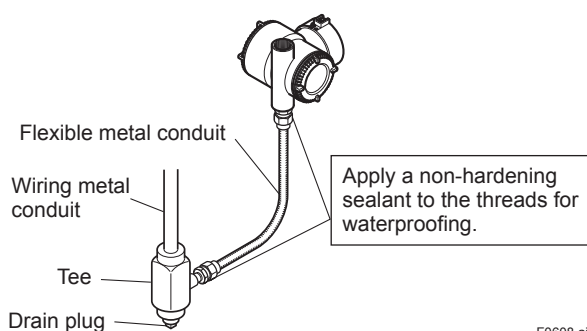


Figure 6.13 Typical Wiring Using Flexible Metal Conduit

(2) Flameproof Type

Wire cables through a flameproof packing adapter, or use a flameproof metal conduit.

- Wiring cable through flameproof packing adapter.

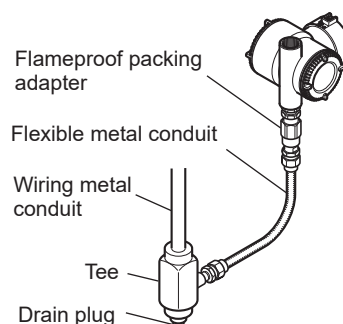


Figure 6.14 Typical Cable Wiring Using Flameproof Packing Adapter

- Flameproof metal conduit wiring

- A seal fitting must be installed near the terminal box connection port for a sealed construction.
- Apply a non-hardening sealant to the threads of the terminal box connection port, flexible metal conduit and seal fitting for waterproofing.

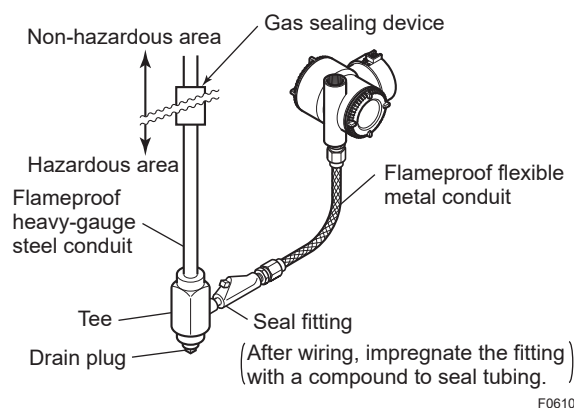


Figure 6.15 Typical Wiring Using Flameproof Metal Conduit



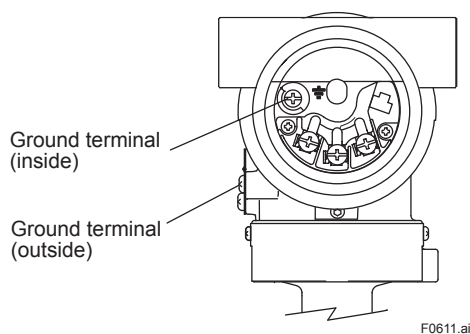
IMPORTANT

When electrical connection code “7” or “C” is specified, the blanking element at the electrical connection port is provided with sealing tape applied. As using sealing tape is prohibited for Ex “d” entry devices by the requirements of IEC 60079-14, remove the sealing tape and apply appropriate seal to the blanking element in accordance with IEC 60079-14, if the instrument is used as Ex “d” equipment.

6.5 Grounding

Grounding is always required for the proper operation of transmitters. Follow the domestic electrical requirements as regulated in each country. For a transmitter with a built-in lightning protector, grounding should satisfy ground resistance of 10Ω or less.

Ground terminals are located on the inside and outside of the terminal box. Either of these terminals may be used.



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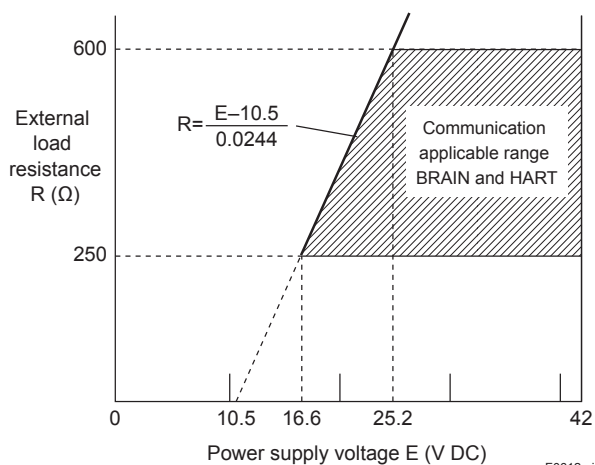
Figure 6.16 Ground Terminals

6.6 Power Supply Voltage and Load Resistance

For 4 to 20 mA output only.

When configuring the loop, make sure that the external load resistance is within the range in the figure below.

(Note) In case of an intrinsically safe transmitter, external load resistance includes safety barrier resistance.



F0612.ai

Figure 6.17 Relationship between Power Supply Voltage and External Load Resistance

7. Operation

7.1 Preparation for Starting Operation

This section describes the operation procedure for the EJ□110□, EJ□120□ and EJ□130□ as shown in figure 7.1a (vertical impulse piping type, high-pressure connection: right side) when measuring the liquid flow rate, and EJ□430□, EJ□440□ and EJ□310□ as shown in figure 7.1b when measuring pressure.



NOTE

Check that the process pressure tap valves, drain valves, and 3-valve manifold stop valves on both the low pressure and high pressure sides are closed, and that the 3-valve manifold equalizing valve is opened.

- (a) Follow the procedures below to introduce process pressure into the impulse piping and transmitter.

■ Differential Pressure Transmitters

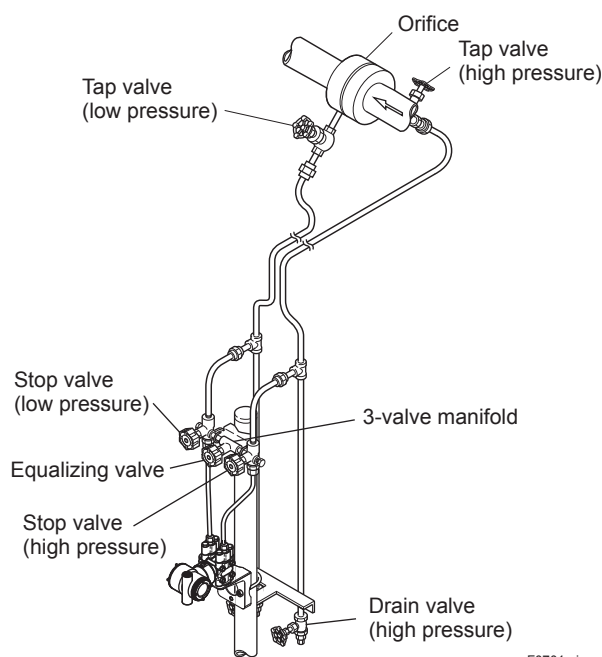
- 1) Open the low pressure and high pressure tap valves to fill the impulse piping with process liquid.
- 2) Slowly open the high pressure stop valve to fill the transmitter pressure-detector section with process liquid.
- 3) Close the high pressure stop valve.
- 4) Gradually open the low pressure stop valve and completely fill the transmitter pressure-detector section with process liquid.
- 5) Close the low pressure stop valve.
- 6) Gradually open the high pressure stop valve. At this time, equal pressure is applied to the low and high pressure sides of the transmitter.
- 7) Check that there are no liquid leaks in the impulse piping, 3-valve manifold, transmitter, or other components.

■ Gauge/Absolute Pressure Transmitters

- 1) Open the tap valve (main valve) to fill the impulse piping with process fluid.
- 2) Gradually open the stop valve to introduce process fluid into the transmitter pressure-detector section.
- 3) Confirm that there is no pressure leak in the impulse piping, transmitter, or other components.

Venting Gas from the Transmitter Pressure-detector Section

- Since the piping in the example of figure 7.1 is constructed to be self-venting, no venting operation is required. If it is not possible to make the piping self-venting, refer to subsection 7.5 for instructions. Leave the equalizing valve open even after venting gas.
- (b) Turn ON power and connect the communicator. Open the terminal box cover, and connect the communicator to the SUPPLY + and – terminals.
 - (c) Using the communicator, confirm that the transmitter is operating properly. Check parameter values or change the setpoints as necessary.
See IM 01C25T03-01E (BRAIN communication) or IM 01C25T01-06EN (HART communication) for communicator operation. If the transmitter is equipped with an integral indicator, its indication can be used to confirm that the transmitter is operating properly.



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Figure 7.1a Liquid Flow Measurement

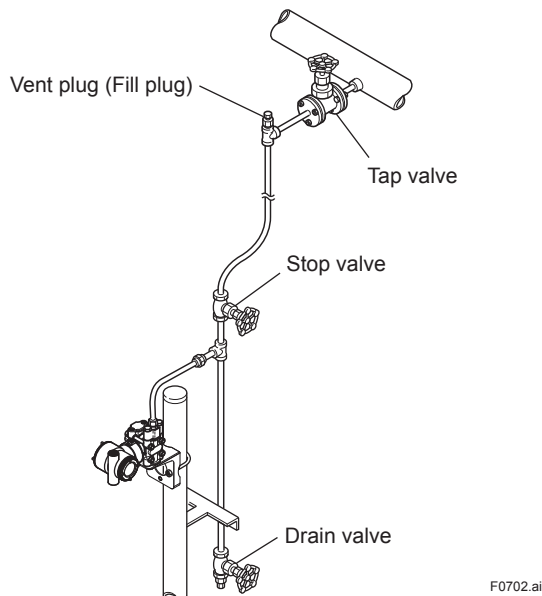


Figure 7.1b Liquid Pressure Measurement

■ Confirming that Transmitter is Operating Properly

Using the BT200



IMPORTANT

- Analog output may change temporally in connecting with BRAIN terminal due to an initial current flowed to it. To prevent communication signal affecting the upper system, it is recommended to install a low-pass filter (approximately 0.1s).
 - Communication signal is superimposed on analog output signal. It is recommended to set a low-pass filter (approximately 0.1s) to the receiver in order to reduce the output effect from communication signal. Before online-communication, confirm that communication signal does not give effect on the upper system.
-
- If the wiring system is faulty, 'communication error' appears on the display.
 - If the transmitter is faulty, 'SELF CHECK ERROR' appears on the display.

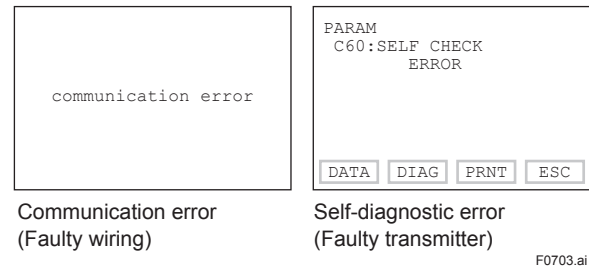


Figure 7.2 BT200 Display

Using the integral indicator

- If the wiring system is faulty, the display stays blank.
- If the transmitter is faulty, an error code is displayed.



Self-diagnostic error on the integral indicator (Faulty transmitter)

F0704.ai

Figure 7.3 Integral Indicator with Error Code



NOTE

If any of the above errors are indicated on the display of the integral indicator or the communicator, refer to subsection 8.5.3 for the corrective action.

■ Verify and Change Transmitter Parameter Setting and Values

The parameters related to the following items are set at factory as specified in order.

- Calibration range
- Integral indicator display
- Output mode
- Software damping (optional)

Other parameters like following are shipped with the default setting.

- Low-cut
- Process alarm setting
- Static pressure range
- Signal characterizer
- Write protection

To confirm or change the values, see IM 01C25T01-06EN or 01C25T03-01E.

7.2 Zero Point Adjustment

After completing preparations for operating the transmitter, adjust the zero point.

Zero point adjustment can be done by turning the transmitter's zero-adjustment screw or by using the communicator. This section describes the procedure for the zero-adjustment screw. For the communicator procedure, see the communication manual.



IMPORTANT

- Do not turn off the power to the transmitter immediately after performing a zero point adjustment. Powering off within 30 seconds of performing this procedure will return the zero point to its previous setting.
- Using zero-point screw requires the parameter EXT_ZERO_ADJUSTMENT is set to "Enabled." When option code /CJ or /CK is specified, the parameter is set to disabled upon shipment. Refer to each communication manual for confirming or changing the setting.



NOTE

Before performing this adjustment, make sure that the external zero adjustment function has NOT been disabled by a parameter setting.



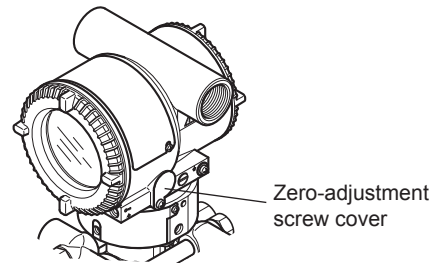
NOTE

- Because the output value may be affected by the transmitter's mounting position, zero adjustment should be executed after installing the transmitter. If you have to adjust zero point before installation, make the posture of the transmitter close to the actual installation state when adjusting zero.
- The factory adjusts the zero point of the absolute pressure transmitters (EJX310A and EJA310E) in the position (horizontal or vertical piping) as specified in the order.

To check the output signal, use a digital multimeter, calibrator, or communicator.

7.2.1 Adjusting Zero Point for Differential Pressure Transmitters

Before adjusting zero point, make sure that the equalizing valve is open.



F0705.ai

Figure 7.4 External Zero Adjustment Screw

The zero-adjustment screw is located inside the cover.

Use a slotted screwdriver to turn the zero-adjustment screw. Equalize the transmitter, then turn the screw clockwise to increase the output or counterclockwise to decrease the output. The zero point adjustment can be made with a resolution of 0.01% of the setting range. The degree of zero adjustments varies with the screw turning speed; turn the screw slowly to make a fine adjustment, quickly to make a rough adjustment.

When using differential pressure transmitters for level measurement and if you cannot obtain the lower range value from the actual measurement value of 0%, refer to subsection 7.2.2(2).

7.2.2 Adjusting Zero Point for Gauge/Absolute Pressure Transmitters

(1) When you can obtain the Low Range Value from the actual measured value of 0% (0 kPa, atmospheric pressure);

For pressure measurement using gauge pressure transmitters, follow the steps below before performing zero point adjustment.

- 1) Close the tap valve (main valve).
- 2) Loosen the fill plug so that the pressure applied to the transmitter is only the head of the seal liquid.
- 3) Adjust the zero point at this status.
- 4) After the adjustment, close the fill plug and then gradually open the tap valve.

Use a slotted screwdriver to turn the zero-adjustment screw. Turn the screw clockwise to increase the output or counterclockwise to decrease the output. The zero point adjustment can be made with a resolution of 0.01% of the setting range. Since the degree of the zero adjustment varies with the screw turning speed, turn the screw slowly to make a fine adjustment and quickly to make a rough adjustment.

(2) When you cannot obtain the Low Range Value from the actual measured value of 0%;

Adjust the transmitter output to the actual measured value obtained by a digital manometer or a glass gauge.

[Example]

The measuring range of 50 to 250 kPa; the actual measured value of 130 kPa.

$$\text{Actual measured value} = \frac{130-50}{250-50} \times 100 = 40.0\% \\ (=10.4\text{mA})$$

Turn the screw to match the output signal to the actual measured value.

7.3 Starting Operation

After completing the zero point adjustment, follow the procedures below to start operation. Steps 1) and 2) are specific to the differential pressure transmitters.

- 1) Close the equalizing valve.
- 2) Gradually open the low pressure stop valve. This places the transmitter in an operational condition.
- 3) Confirm the operating status. If the output signal exhibits wide fluctuations (hunting) due to periodic variation in the process pressure, use the communicator to dampen the transmitter output signal. Confirm the hunting using a receiving instrument or the integral indicator, and set the optimum damping time constant.
- 4) After confirming the operating status, perform the following.



IMPORTANT

- Remove the communicator from the terminal box, and confirm that none of the terminal screws are loose.
- Close the terminal box cover and the amplifier cover. Screw each cover in tightly until it will not turn further.
- There are two covers that must be locked on the ATEX Flameproof type transmitters. An Allen head bolt (shrouding bolt) under the edge of each cover is used to lock the cover. When the shrouding bolt is driven counterclockwise with an Allen wrench, the bolt rotates upward and locks the cover. (See section 8.4.) After locking the covers, confirm that they are secure and cannot be opened by hand.
Tighten the zero-adjustment cover mounting screw to secure the cover.
- Tighten the zero-adjustment cover mounting screw to secure the cover.

7.4 Shutting Down the Transmitter

Shut down the transmitter as follows. Steps 2) and 3) are specific to the differential pressure transmitters.

- 1) Turn off the power.
- 2) Close the low pressure stop valve.
- 3) Open the equalizing valve.
- 4) Close the high pressure stop valve.
- 5) Close the high pressure and low pressure tap valves.



NOTE

- Whenever shutting down the transmitter for a long period, remove any process fluid that is in the transmitter pressure-detector section.
- The equalizing valve must be left OPEN.

7.5 Venting or Draining Transmitter Pressure-detector Section

Since this transmitter is designed to be self-draining and self-venting with vertical impulse piping connections, neither draining nor venting will be required if the impulse piping is configured appropriately for self-draining or self-venting operation.

If condensate (or gas) collects in the transmitter pressure-detector section, the measured pressure may be in error. If it is not possible to configure the piping for self-draining (or self-venting) operation, you will need to loosen the drain (vent) screw on the transmitter to completely drain (vent) any stagnated liquid (gas.)

However, since draining condensate or bleeding off gas disturbs the pressure measurement, this should not be done when the loop is in operation.

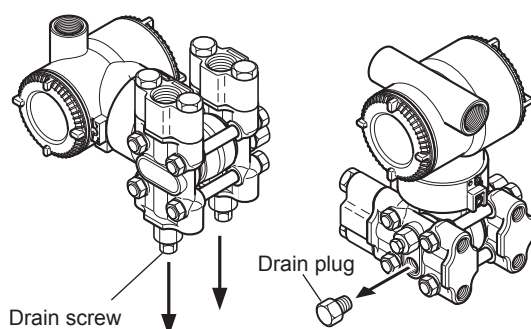


WARNING

Since the accumulated liquid (or gas) may be toxic or otherwise harmful, take appropriate care to avoid contact with the body, or inhalation of vapors.

7.5.1 Draining Condensate

- 1) Gradually open the drain screw or drain plug and drain the transmitter pressure-detector section. (See figure 7.5.)
- 2) When all accumulated liquid is completely removed, close the drain screw or drain plug.
- 3) Tighten the drain screw to a torque of 10 N·m, and the drain plug to a torque of 34 to 39 N·m.



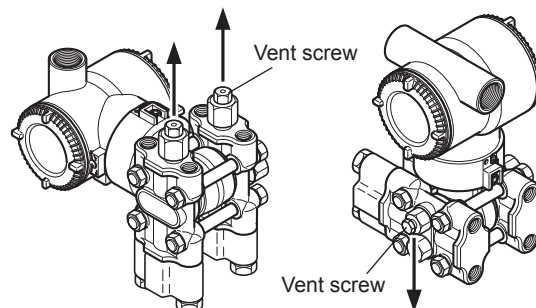
When you loosen the drain screw or drain plug, the accumulated liquid will be expelled in the direction of the arrow.

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Figure 7.5 Draining the Transmitter

7.5.2 Venting Gas

- 1) Gradually open the vent screw to vent gas from the transmitter pressure-detector section. (See figure 7.6.)
- 2) When the transmitter is completely vented, close the vent screw.
- 3) Tighten the vent screw to a torque of 10 N·m.



When you loosen the vent screw, the gas escapes in the direction of the arrow.

F0707.ai

Figure 7.6 Venting the Transmitter

7.6 Local Parameter Setting



WARNING

The local push button on the integral indicator must not be used in a hazardous area. When it is necessary to use the push button, operate it in a non-hazardous location. In the case of the explosion-proof/flameproof type transmitter, please refer to section 2.9 for each product.

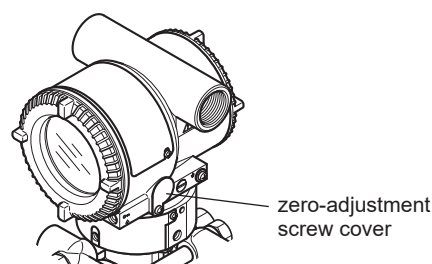


IMPORTANT

- Do not turn off the power to the transmitter immediately after performing parameter setting. Powering off within 30 seconds of performing this procedure will return the parameter to its previous setting.
- The parameter 'Ext SW(HART)' or 'J55:EXT ZERO ADJ (BRAIN)' must be "Enabled" to perform this configuration. When option code /CJ or /CK is specified, the parameter is set to disabled upon shipment. Please refer to either IM 01C25T01-06EN for the HART communication type, or IM 01C25T03-01E for the BRAIN communication type for the setting procedure.
- The Local Parameter Setting function is available with HART or BRAIN communication type.
LCD update will be slower at low ambient temperature, and it is recommended to use LPS function at temperatures above -10 degrees C.

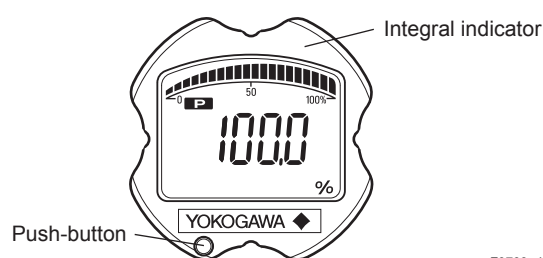
7.6.1 Local Parameter Setting (LPS) Overview

Parameter configuration by the zero-adjustment screw and push button (integral indicator code E) offers easy and quick setup for parameters of Loop test, Tag number, Unit, LRV, URV, Damping, Output mode (linear/square root), Display out 1, and Re-range by applying actual pressure (LRV/URV) and Device Information. There is no effect on measurement signal (analog output or communication signal) when Local Parameter Setting is carried out.



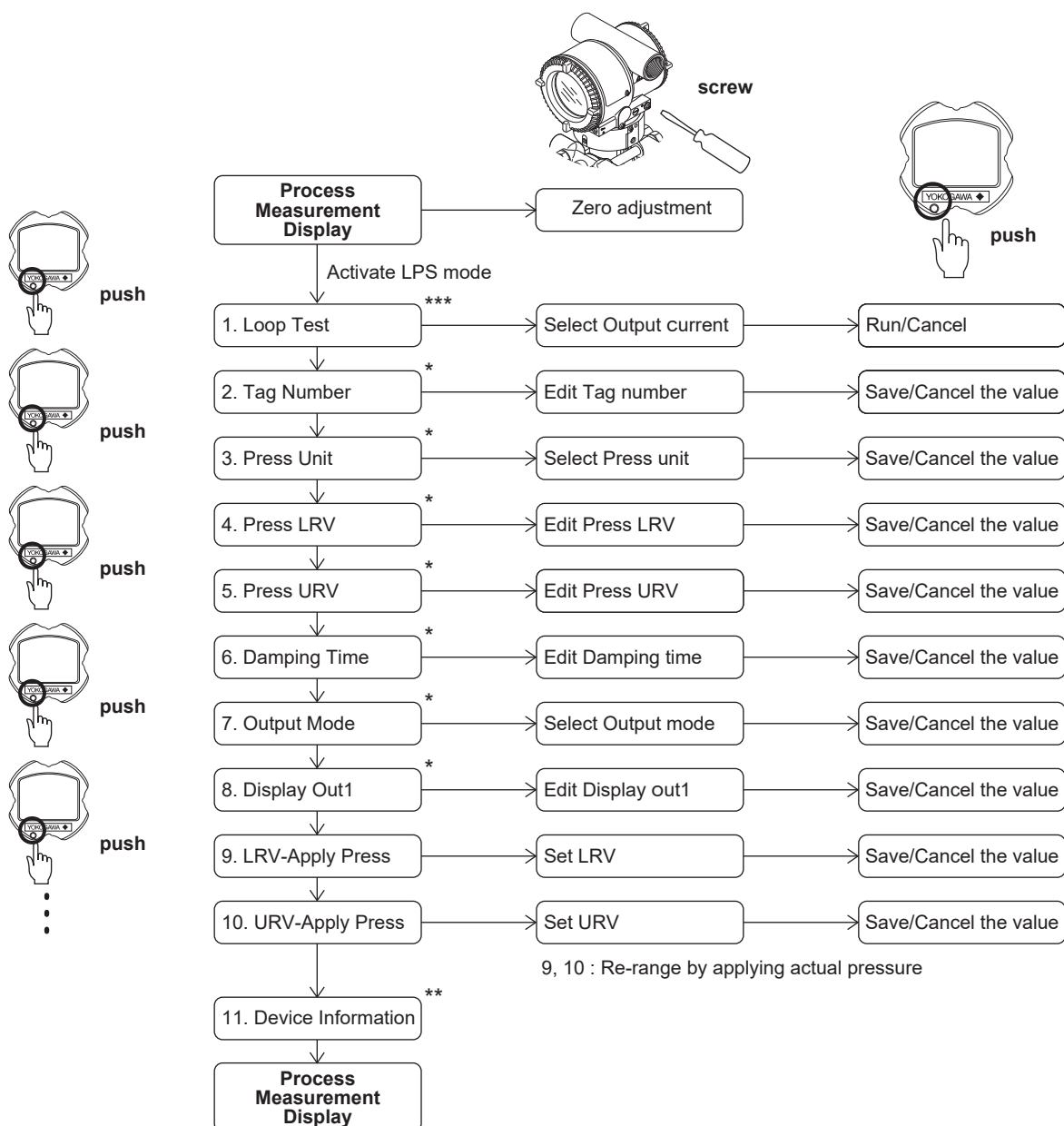
F0708.ai

Figure 7.7 Zero-adjustment Screw



F0709.ai

Figure 7.8 Range-Setting Switch (push button)



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*: These parameter configurations are available with the software revision (SOFT REV)2.03 or later.

Software revision can be checked via a field communicator (HART/BRAIN) or DTM. Please refer to IM 01C25T01 for HART parameter "Software rev" and IM 01C25T03 for BRAIN "SOFT REV."

**: This parameter configuration is available with the software revision (SOFT REV) 3.01 or later.

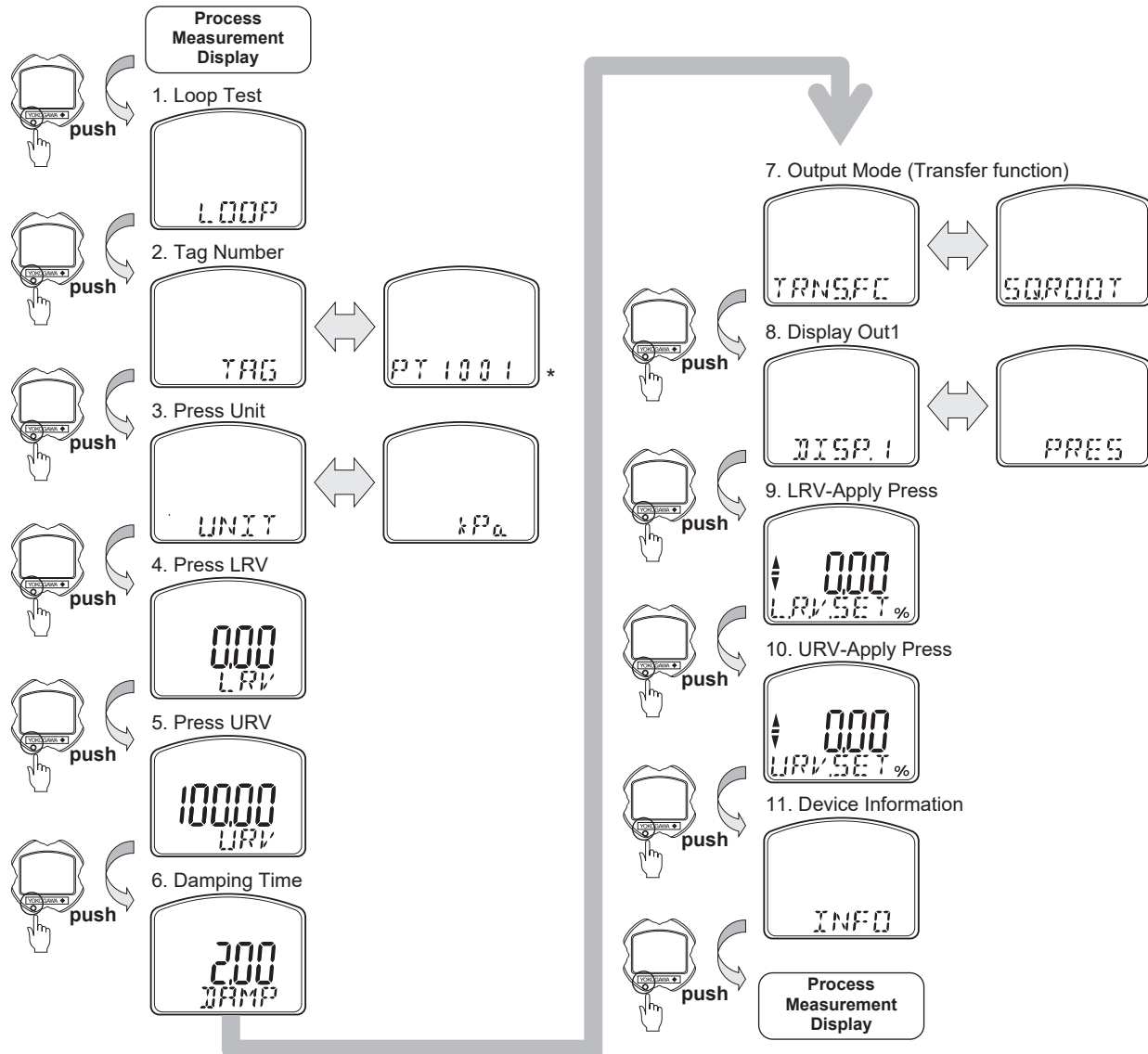
***: This parameter configuration is available with the software revision (SOFT REV) 5.01 or later.

7.6.2 Activating Local Parameter Setting

Press the push button on the integral indicator to activate the Local Parameter Setting mode. The transmitter will exit automatically from the Local Parameter Setting mode if no operation is carried out for 10 minutes.

7.6.3 Parameter Setting Review

Current setting value for the below parameters are shown sequentially by each press of the push button. Loop test, Tag number, Unit, LRV, URV, Damping, Output mode (linear/square root), Display out 1, Re-range by applying actual pressure (LRV/URV) and Device Information.



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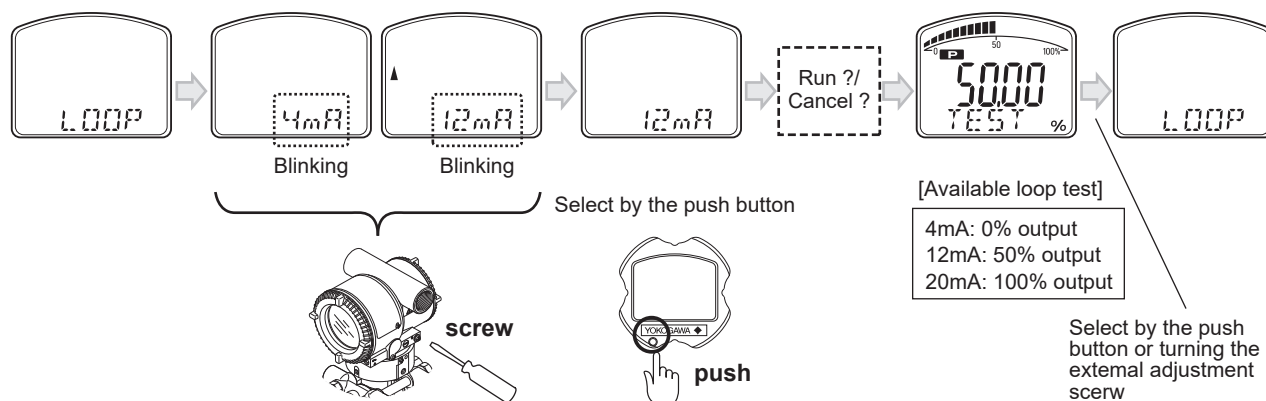
*: [Software revision: 2.03] Up to 6 characters are shown in the bottom line. To review after the 7th character, turn the zero-adjustment screw and press the push button six times or more.
[Software revision: 3.01 or later] The display automatically scrolls to show all the characters.

To configure each parameter value, turn the zero-adjustment screw on each parameter screen after activating the Local Parameter Setting mode.

To cancel the Local Parameter Setting configuration, please refer to 7.6.13 Save or Cancel and 7.6.14 Abort Configuration.

7.6.4 Loop Test Configuration

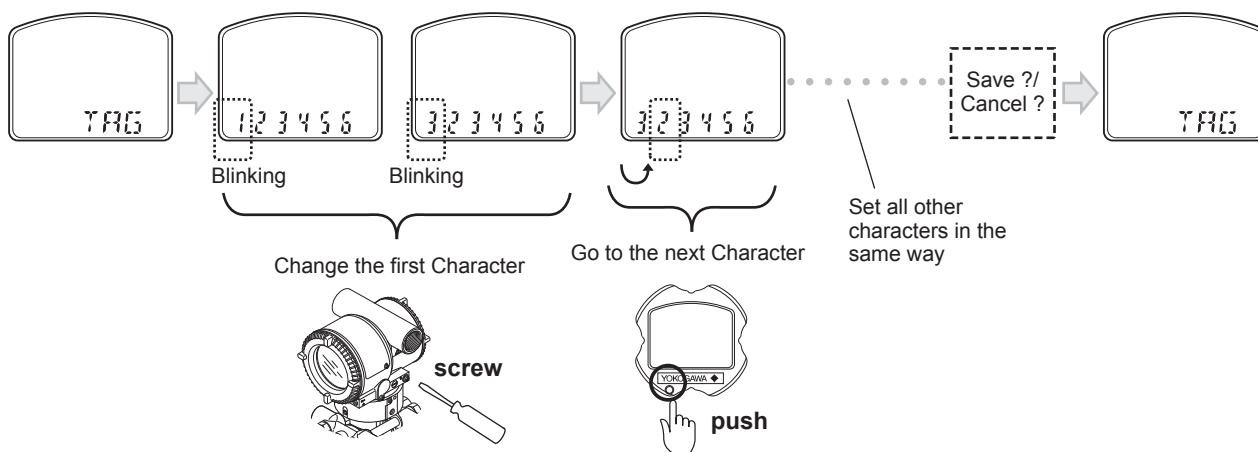
Loop test configuration can be changed as below. Output current value (4 mA, 12 mA or 20 mA) for Loop test can be selected by turning the zero-adjustment screw.



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7.6.5 Tag Number Configuration

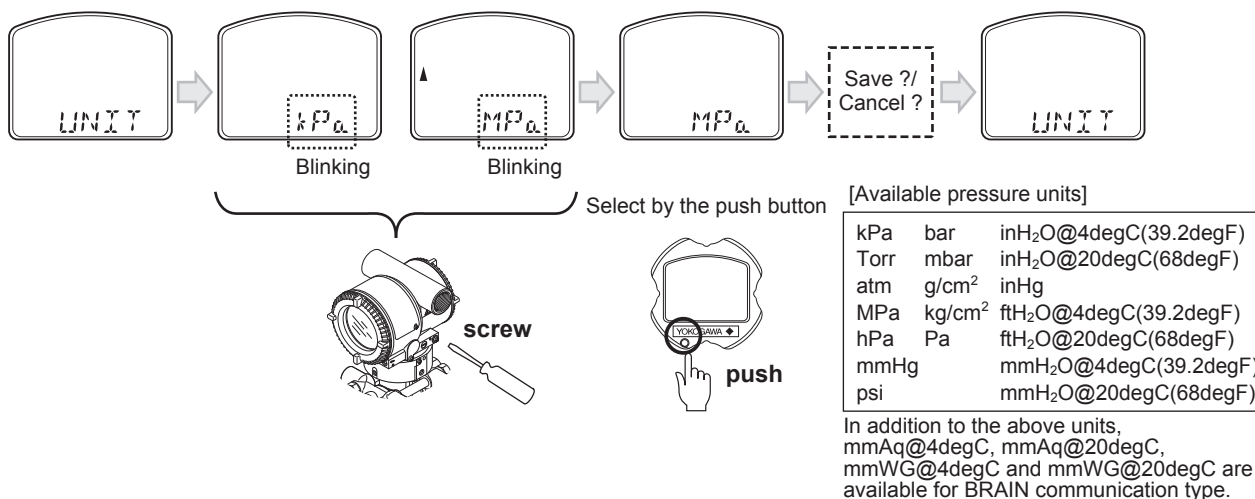
Tag Number is edited by turning the zero-adjustment screw. Up to 8 alphanumeric characters for HART or 16 alphanumeric characters for BRAIN can be set.



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7.6.6 Pressure Unit Configuration

Pressure unit for the below table can be changed as below. By turning the zero-adjustment screw, user can scroll between the various available pressure units.



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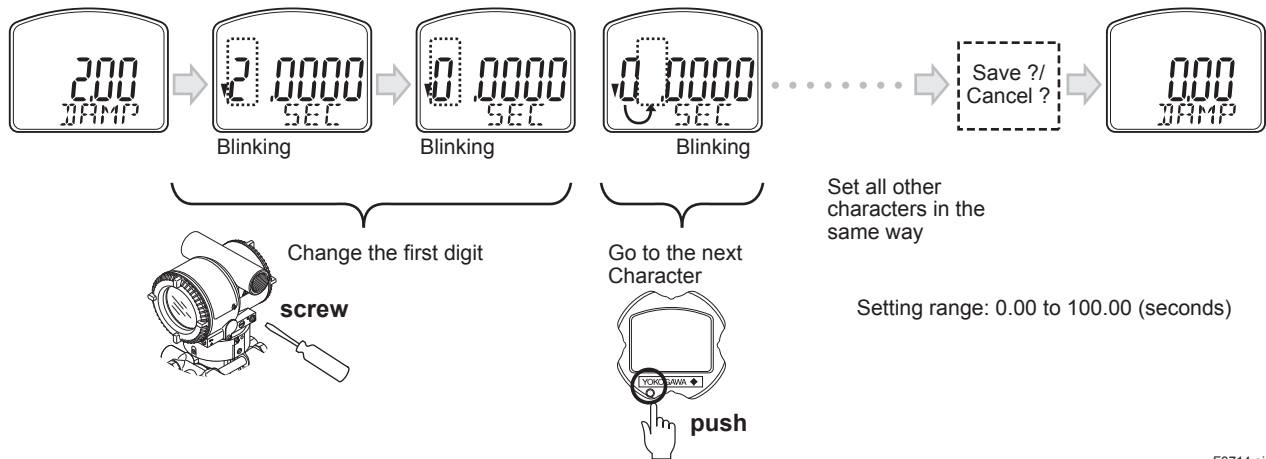
7.6.7 Pressure LRV/URV Configuration

Pressure LRV and URV can be set. The number for each digit is changed by turning the zero-adjustment screw and set by pressing the push button. Please refer to 7.6.8 Damping Time Constant Configuration for how to change the numerical value.

When the setting is out of the limit, an alarm will be generated.

7.6.8 Damping Time Constant Configuration

The damping time constant for the amplifier assembly can be set. Quick Response Parameter is automatically set to ON when the damping time constant is set to less than 0.5 seconds. Damping time constant is rounded off to two decimal places.



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7.6.9 Output Mode Configuration

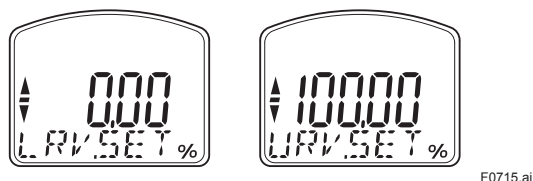
Pressure Output Mode (Linear or Sq root) can be selected by turning the zero-adjustment screw. Please refer to 7.6.6 Pressure Unit Configuration for how to select and set the enumerated value.

7.6.10 Display Out 1 Configuration

Display Out1 can be selected by turning the zero-adjustment screw. Please refer to 7.6.6 Pressure Unit Configuration for how to select and set the enumerated value.

7.6.11 Re-range by applying actual pressure (LRV/URV).

This feature allows the lower and upper range values to be setup with the actual input applied.



Follow the procedure below to change the LRV and URV settings.

[Example]

Rerange LRV to 0 and URV to 3 MPa.

- 1) Connect the transmitter and apparatus as shown in Figure 8.1 and warm it up for at least five minutes.
- 2) Press the push-button. The integral indicator then displays "LRV.SET."
- 3) Apply a pressure of 0 kPa (atmospheric pressure) to the transmitter. (Note 1)
- 4) Turn the zero-adjustment screw in the desired direction. The integral indicator displays the output signal in %. (Note 2)
- 5) Adjust the output signal to 0% (1 V DC) by rotating the zero-adjustment screw. Press the push button to save the value. Doing so completes the LRV setting. (Note 3)
- 6) Press the push-button. The integral indicator then displays "URV.SET."
- 7) Apply a pressure of 3 MPa to the transmitter. (Note 1)
- 8) Turn the zero-adjustment screw in the desired direction. The integral indicator displays the output signal in %. (Note 2)
- 9) Adjust the output signal to 100% (5 V DC) by rotating the zero-adjustment screw. Press the button to save the value. Doing so completes the URV setting.
- 10) Press the push-button. The transmitter then switches back to the normal operation mode with the measurement range of 0 to 3 MPa.

Note 1: Wait until the pressure inside the pressure-detector section has stabilized before proceeding to the next step.

Note 2: If the pressure applied to the transmitter exceeds the previous LRV (or URV), the integral indicator may display error number "AL.30" (In this case, the output signal percent and "AL.30" are displayed alternately every two seconds). Although "AL.30" is displayed, you may proceed to the next step. However, should any other error number be displayed, take the appropriate measure in reference to, "Errors and Countermeasures" in each communication manual.

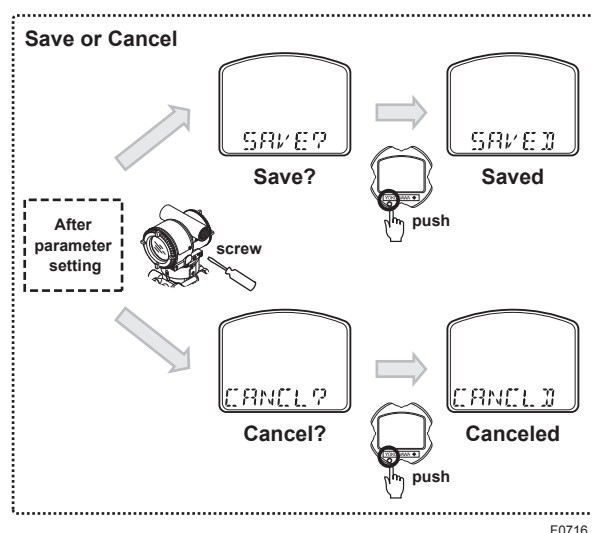
Note 3: Changing the lower range value (LRV) also automatically changes the upper range value (URV), keeping the span constant. New URV=previous URV+(new LRV–previous LRV)

7.6.12 Device Information Display

Model name, communication protocol, device revision and software revision are displayed repeatedly.

7.6.13 Save or Cancel

At the end of each parameter setting, select "Save" or "Cancel" by the zero-adjustment screw and press the push button to save or cancel the configuration.

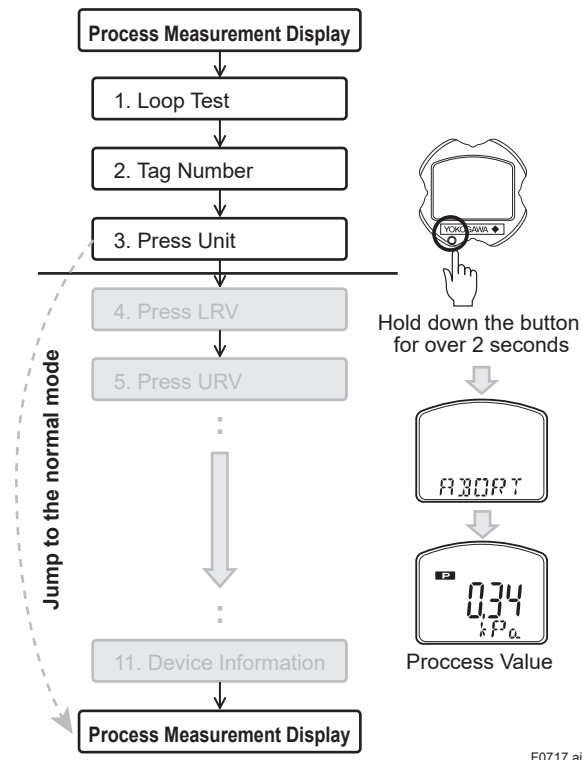


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7.6.14 Abort Configuration

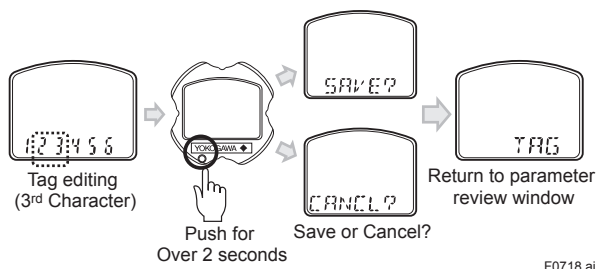
7.6.14.1 Abort Configuration (Menu)

Hold down the push button for over 2 seconds to exit the Local Parameter Setting mode.



7.6.14.2 Abort Configuration (Parameter)

To exit the configuration while editing the value, hold down the push button for over 2 seconds and select "Save" or "Cancel."



7.6.15 Local Parameter Setting Lock

To disable parameter changes by the Local Parameter Setting there are three different ways.

	Locked features
Communication Parameter Ext SW =disable (EXT ZERO ADJ = disable)	- External Zero Adjustment - Local Parameter Setting
Communication Parameter Write Protect = On (WRT PROTECT = Yes)	- Local Parameter Setting - All Communication Parameters *
Hardware write protection switch on CPU assembly = D (Disable)	- Local Parameter Setting - All Communication Parameters *

*External Zero Adjustment is unlocked.

The above parameter setting is carried out by using field communicator or DTM. See the user's manual IM 01C25T (HART/BRAIN) for the setting procedure.

Reviewing local parameter setting by push button on the integral indicator is available at any time even when the Local Parameter Setting is locked.

7.6.16 Others

- Difference between BRAIN and HART

	HART	BRAIN
Tag Number	Up to 8 characters can be set	Up to 16 characters can be set
Output Mode (Linear/Sq root)	"TRNS.FC" shown on the integral indicator	"OUT.MOD" shown on the integral indicator

- The degree of adjustment depends on the speed of turning the adjustment screw. Turn the screw slowly for fine tuning and turn the screw fast for coarse tuning.

8. Maintenance

8.1 Overview



WARNING

Since the accumulated process fluid may be toxic or otherwise harmful, take appropriate care to avoid contact with the body or inhalation of vapors when draining condensate or venting gas from the transmitter pressure-detector section and even after dismounting the instrument from the process line for maintenance.

Maintenance of the transmitter is easy due to its modular construction. This chapter describes the procedures for calibration, adjustment, and the disassembly and reassembly procedures required for component replacement.

Transmitters are precision instruments. Please carefully and thoroughly read the following sections for information on how to properly handle them while performing maintenance.



IMPORTANT

- As a rule, maintenance of this transmitter should be done in a shop that has all the necessary tools.
- The CPU assembly contains sensitive parts that can be damaged by static electricity. Take precautions such as using a grounded wrist strap when handling electronic parts or touching the board circuit patterns. Also be sure to place the removed CPU assembly into a bag with an antistatic coating.

8.2 Calibration Instruments Selection

Table 8.1 lists the instruments that can be used to calibrate a transmitter. When selecting an instrument, consider the required accuracy level. Exercise care when handling these instruments to ensure they maintain the specified accuracy.

8.3 Calibration

Use the procedure below to check instrument operation and accuracy during periodic maintenance or troubleshooting.

- 1) Connect the instruments as shown in figure 8.1 and warm up the instruments for at least five minutes (ten minutes for EJ□310□.)



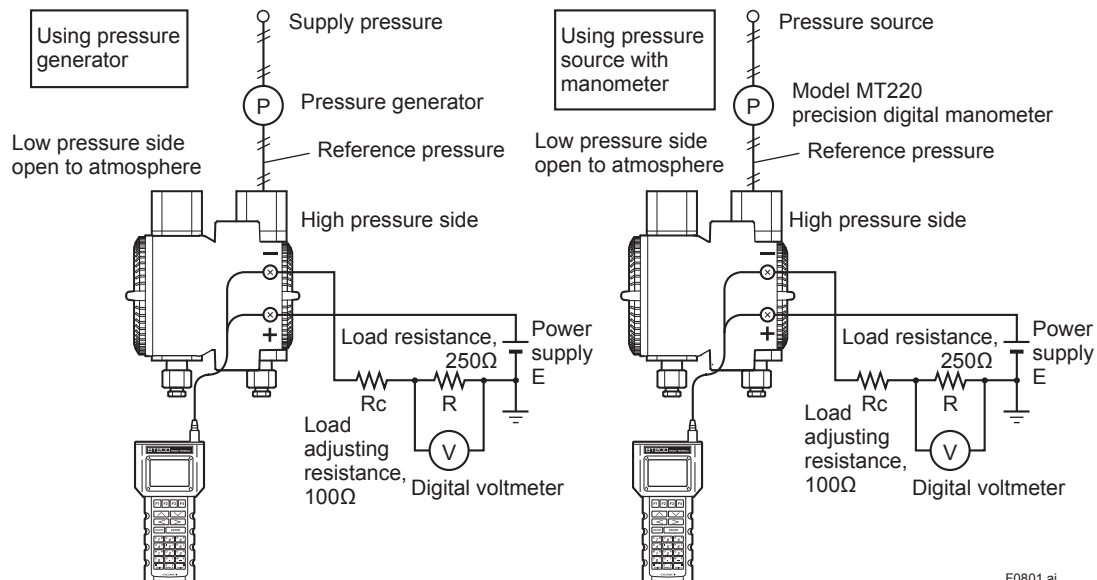
IMPORTANT

- To adjust the transmitter for highest accuracy, make adjustments with the power supply voltage and load resistance including leadwire resistances set close to the conditions under which the transmitter is installed.
 - If the measurement range 0% point is 0 kPa or shifted in the positive direction (suppressed zero), the reference pressure should be applied as shown in the figure. If the measurement range 0% point is shifted in the negative direction (elevated zero), the reference pressure should be applied using a vacuum pump.
- 2) Apply reference pressures of 0%, 50%, and 100% of the measurement range to the transmitter. Calculate the errors (differences between digital voltmeter readings and reference pressures) as the pressure is increased from 0% to 100% and is decreased from 100% to 0%, and confirm that the errors are within the required accuracy.

Table 8.1 Instruments Required for Calibration

Name	Yokogawa-recommended Instrument	Remarks
Power supply	Model SDBT or SDBS distributor	4 to 20 mA DC signal
Load resistor	Model 2792 standard resistor [250 Ω $\pm$ 0.005%, 3 W] Load adjustment resistor [100 Ω $\pm$ 1%, 1 W]	
Voltmeter	Model 2501 A digital multimeter Accuracy (10V DC range): $\pm$ (0.002% of rdg + 1 dgt)	
Digital manometer	Model MT220 precision digital manometer 1) For 10 kPa class Accuracy: $\pm$ (0.015% of rdg + 0.015% of F.S.) for 0 to 10 kPa $\pm$ (0.2% of rdg + 0.1% of F.S.) for -10 to 0 kPa 2) For 130 kPa class Accuracy: $\pm$ 0.02% of rdg for 25 to 130 kPa $\pm$ 5digits for 0 to 25 kPa $\pm$ (0.2% of rdg + 0.1% of F.S.) for -80 to 0 kPa 3) For 700 kPa class Accuracy: $\pm$ (0.02% of rdg + 3digits) for 100 to 700 kPa $\pm$ 5 digits for 0 to 100 kPa $\pm$ (0.2% of rdg + 0.1% of F.S.) for -80 to 0 kPa 4) For 3000 kPa class Accuracy: $\pm$ (0.02% of rdg + 10 digits) for 0 to 3000 kPa $\pm$ (0.2% of rdg + 0.1% of F.S.) for -80 to 0 kPa 5) For 130 kPa abs class Accuracy: $\pm$ (0.03% of rdg + 6 digits) for 0 to 130 kPa abs	Select a manometer having a pressure range close to that of the transmitter.
Pressure generator	Model 7674 pneumatic pressure standard for 200 kPa {2 kgf/cm ² }, 25 kPa {2500 mmH ₂ O} Accuracy: $\pm$ 0.05% of F.S. Dead weight gauge tester 25 kPa {2500 mmH ₂ O} Accuracy: $\pm$ 0.03% of setting	Requires air pressure supply. Select the one having a pressure range close to that of the transmitter.
Pressure source	Model 6919 pressure regulator (pressure pump) Pressure range: 0 to 133 kPa {1000 mmHg}	Prepare the vacuum pump for negative pressure ranges.

Note: The above table contains the instruments capable of performing calibration to the 0.2% level. Since special maintenance and management procedures involving traceability of each instrument to higher-level standards are required for calibration to the 0.1% or higher level, there may be difficulties in calibration to this level in the field. For calibration to the 0.1% level, contact Yokogawa representatives from which the instrument was purchased or the nearest Yokogawa office.



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Figure 8.1 Instrument Connections (EJX110A, 4 to 20mA output)

8.4 Disassembly and Reassembly

This section describes procedures for disassembly and reassembly for maintenance and component replacement.

Always turn OFF power and shut off and release pressures before disassembly. Use proper tools for all operations. Table 8.2 shows the tools required.

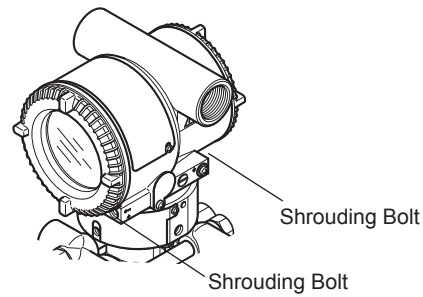
Table 8.2 Tools for Disassembly and Reassembly

Tool	Quantity	Remarks
Phillips screwdriver	1	JIS B4633, No. 2
Slotted screwdriver	1	
Allen wrenches	3	JIS B4648 One each, nominal 2.5, 3, 4 and 6 mm Allen wrenches ASME B18.3. One nominal 1/4 Allen wrench (for 1/4NPT bug screen)
Wrench	1	Width across flats, 17 mm
Torque wrench	1	
Adjustable wrench	1	
Socket wrench	1	Width across flats, 16 mm
Socket driver	1	Width across flats, 5.5 mm
Tweezers	1	

CAUTION

Precautions for ATEX Flameproof Type Transmitters

- Flameproof type transmitters must be, as a rule, removed to a non-hazardous area for maintenance and be disassembled and reassembled to the original state.
- On the flameproof type transmitters the two covers are locked, each by an Allen head bolt (shrouding bolt). When a shrouding bolt is driven clockwise by an Allen wrench, it is going in and cover lock is released, and then the cover can be opened.
When a cover is closed it should be locked by a shrouding bolt without fail. Tighten the shrouding bolt to a torque of 0.7 N·m.



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Figure 8.2 Shrouding Bolts

8.4.1 Replacing the Integral Indicator

CAUTION

Cautions for Explosion-proof/Flameproof Type Transmitters

Users are prohibited by law from modifying the construction of a explosion-proof/flameproof type transmitter. This would invalidate the agency approval for the use of the transmitter in a rated area.

It follows that the user is prohibited from using a explosion-proof/flameproof type transmitter with its integral indicator removed, or from adding an integral indicator to a transmitter. If such modification is absolutely required, contact Yokogawa.

This subsection describes the procedure for replacing an integral indicator. (See figure 8.3)

■ Removing the Integral Indicator

- 1) Remove the cover.
- 2) While supporting the integral indicator with one hand, loosen its two mounting screws.
- 3) Dismount the LCD board assembly from the CPU assembly.
When doing this, carefully pull the LCD board assembly straight forward so as not to damage the connector pins between it and the CPU assembly.

■ Attaching the Integral Indicator

- 1) Align both the LCD board assembly and CPU assembly connectors and engage them.
- 2) Insert and tighten the two mounting screws.
- 3) Replace the cover.

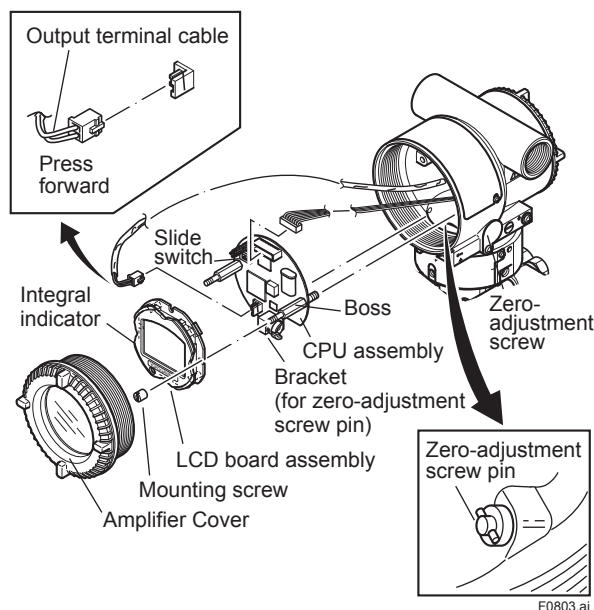


Figure 8.3 Removing and Attaching LCD Board Assembly and CPU Assembly

8.4.2 Replacing the CPU Board Assembly

This subsection describes the procedure for replacing the CPU assembly. (See figure 8.3)

■ Removing the CPU Assembly

- 1) Remove the cover. If an integral indicator is mounted, refer to subsection 8.4.1 and remove the indicator.
- 2) Turn the zero-adjustment screw to the position (where the screw head slot is horizontal) as shown in figure 8.3.

- 3) Disconnect the output terminal cable (cable with brown connector at the end). When doing this, lightly press the side of the CPU assembly connector and pull the cable connector to disengage.
- 4) Use a socket driver (width across flats, 5.5mm) to loosen the two bosses.
- 5) Carefully pull the CPU assembly straight forward to remove it.
- 6) Disconnect the flat cable (cable with white connector at the end) that connects the CPU assembly and the capsule.



NOTE

Be careful not to apply excessive force to the CPU assembly when removing it.

■ Mounting the CPU Assembly

- 1) Connect the flat cable (with white connector) between the CPU assembly and the capsule.
- 2) Connect the output terminal cable (with brown connector).



NOTE

Make certain that the cables do not get pinched between the case and the edge of the CPU assembly.

- 3) Align and engage the zero-adjustment screw pin with the groove on the bracket on the CPU assembly. Then insert the CPU board assembly straight onto the post in the amplifier case.
- 4) Tighten the two bosses. If the transmitter is equipped with an integral indicator, refer to subsection 8.4.1 to mount the indicator.



NOTE

Confirm that the zero-adjustment screw pin is placed properly in the groove on the bracket prior to tightening the two bosses. If it is not, the zero-adjustment mechanism will be damaged.

- 5) Replace the cover.

8.4.3 Cleaning and Replacing the Capsule Assembly

This subsection describes the procedures for cleaning and replacing the capsule assembly. (See figure 8.4.)



CAUTION

Cautions for Flameproof Type Transmitters

Users are prohibited by law from modifying the construction of a flameproof type transmitter. If you wish to replace the capsule assembly with one of a different measurement range, contact Yokogawa.

The user is permitted, however, to replace a capsule assembly with another of the same measurement range. When doing so, be sure to observe the following.

- The replacement capsule assembly must have the same part number as the one being replaced.
- The section connecting the transmitter and capsule assembly is a critical element in preservation of flameproof performance, and must be checked to verify that it is free of dents, scratches, and other defects.
- After completing maintenance, be sure to securely tighten the setscrews that fasten the transmitter section and pressure-detector section together.

■ Removing the Capsule Assembly



IMPORTANT

Exercise care as follows when cleaning the capsule assembly.

- Handle the capsule assembly with care, and be especially careful not to damage or distort the diaphragms that contact the process fluid.
- Do not use a chlorinated or acidic solution for cleaning.
- Rinse thoroughly with clean water after cleaning.

- 1) Remove the CPU assembly as shown in subsection 8.4.2.
- 2) Remove the two setscrews that connect the transmitter section and pressure-detector section.
- 3) Remove the hexagon-head screw and the stopper.
- 4) Separate the transmitter section and pressure-detector section.
- 5) Remove the nuts from the four flange bolts.
- 6) While supporting the capsule assembly with one hand, remove the cover flange.
- 7) Remove the capsule assembly.
- 8) Clean the capsule assembly or replace with a new one.

■ Reassembling the Capsule Assembly

- 1) Insert the capsule assembly between the flange bolts, paying close attention to the relative positions of the H (high pressure side) and L (low pressure side) marks on the capsule assembly.
Replace the two capsule gaskets with new gaskets.
- 2) Install the cover flange on the high pressure side, and use a torque wrench to tighten the four nuts uniformly to a torque shown below.

Model	EJ□110□, EJ□310□, EJ□430□		EJ□120□
	Wetted parts material code		
	S, L	H,M,T,A,D,B,W	
Torque(N·m) {kgf·m}	17* {1.7}	40 {4.1}	40 {4.1}

*: 40 for measurement span code F, option code /HD or /HD2.

Model	EJ□130□				EJ□440□	
	Bolts & nuts material code					
	G, C	J	K	H	G, C	J
Torque(N·m) {kgf·m}	90 {9.2}	110 {11.2}	210 {21.4}	170 {17.3}	70 {7.1}	100 {10.2}

- 3) After the pressure-detector section has been reassembled, a leak test must be performed to verify that there are no pressure leaks.
- 4) Reattach the transmitter section to the pressure-detector section.
Reattach the stopper with the hexagon-head screw.
- 5) Tighten the two setscrews. (Tighten the screws to a torque of 1.5 N·m)
- 6) Install the CPU assembly according to subsection 8.4.2.
- 7) After completing reassembly, adjust the zero point and recheck the parameters.

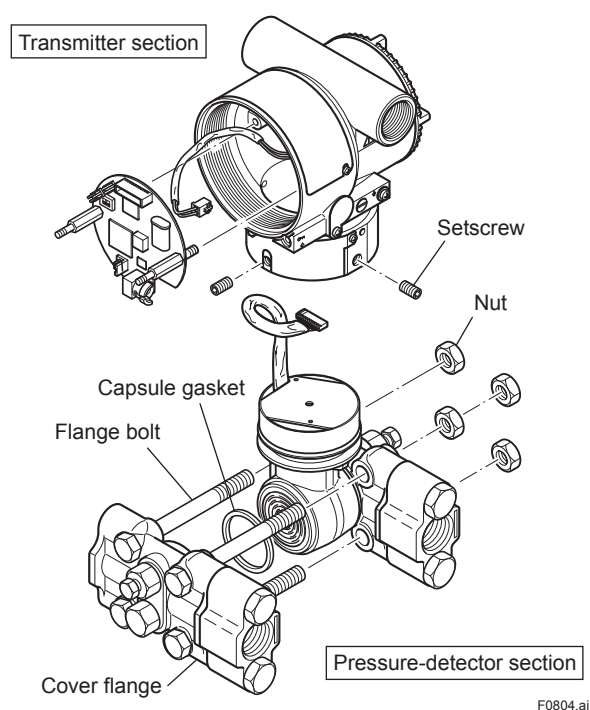


Figure 8.4 Removing and Mounting the Pressure-detector Section



IMPORTANT

If you remove the vent/drain plugs when disassembling EJ□130□ and EJ□440□ transmitters, install the plugs again before reassembling the cover flanges with the capsule assembly.

8.4.4 Replacing the Process Connector Gaskets

This subsection describes process connector gasket replacement. (See figure 8.5.)

- Loosen the two bolts, and remove the process connectors.
- Replace the process connector gaskets.
- Remount the process connectors. Tighten the bolts securely and uniformly to a torque shown below, and verify that there are no pressure leaks.

Model	EJ□110□ EJ□120□ EJ□130□ EJ□310□ EJ□430□	EJ□440□	
		C capsule	D capsule
Torque(N·m) {kgf·m}	39 to 49 {4 to 5}	49 to 59 {5 to 6}	

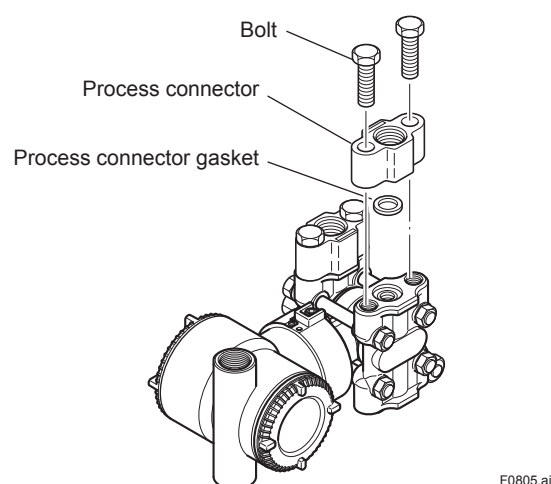


Figure 8.5 Removing and Mounting the Process Connector

8.4.5 Cleaning/replacing the bug screen (Option Code /BS)

The procedures for cleaning and replacing the bug screen are described below. (See figure 8.6)

- Loosen the bug screen (one screw) with a Allen wrench (JIS B4648 Nominal 6 or ASME B18.3 Nominal 1/4 *1 *2).
- Clean or replace the bug screen. (Tighten with a torque of approximately 3 N·m)

*1: Use an allen wrench (JIS B4648 No. 6) for EJ□110□, EJ□120□, EJ□430□, EJ□440□ with option code /BS and process connection code 0, 1, 2, C or Q.

*2: Use an allen wrench (ASME B18.3 Nominal 1/4). for EJ□110□, EJ□120□, EJX430□, EJX440□ with option code /BS and Process connection code 3, 4, 5, D, R, B or G.

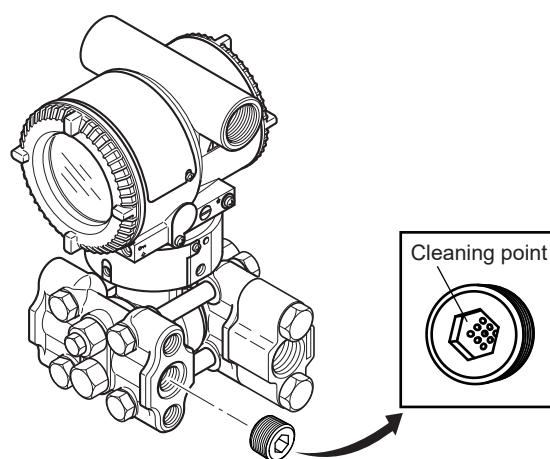


Figure 8.6 Remove/Install a bug screen



IMPORTANT

Keep the following in mind when cleaning the bug screen:

- Do not use chlorine or acidic solutions for cleaning.
- After cleaning, rinse with clean water and dry until the water is completely drained.

8.5 Troubleshooting

If any abnormality appears in the measured values, use the troubleshooting flow chart below to isolate and remedy the problem. Since some problems have complex causes, these flow charts may not identify all. If you have difficulty isolating or correcting a problem, contact Yokogawa service personnel.

8.5.1 Basic Troubleshooting

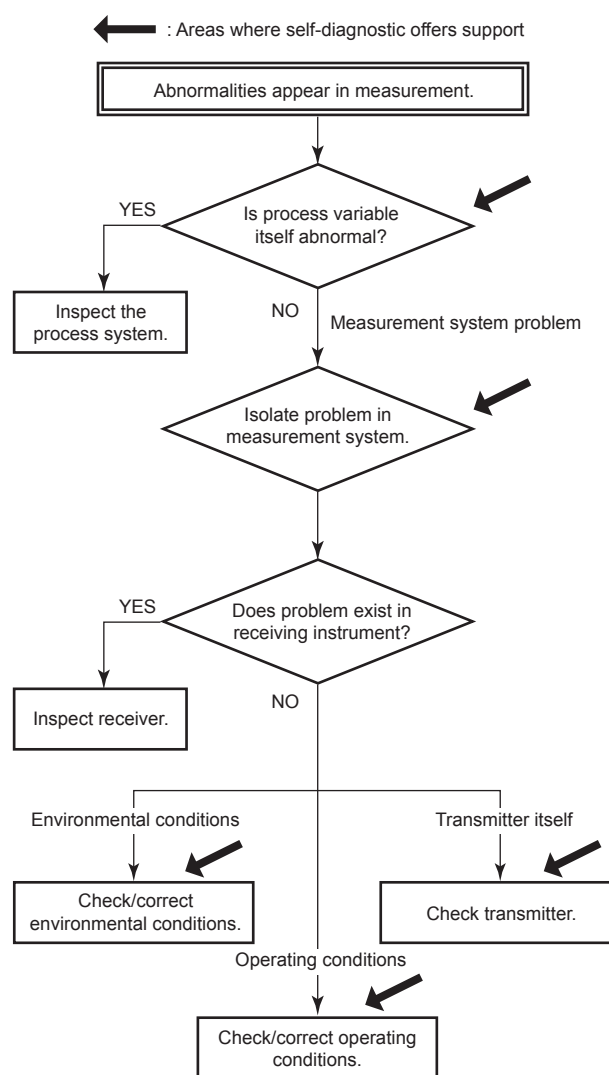
First determine whether the process variable is actually abnormal or a problem exists in the measurement system.

If the problem is in the measurement system, isolate the problem and decide what corrective action to take.

This transmitter is equipped with a self-diagnostic function which will be useful in troubleshooting, and the transmitter equipped with an integral indicator will show an alarm code as a result of self-diagnosis.

See subsection 8.5.3 for the list of alarms.

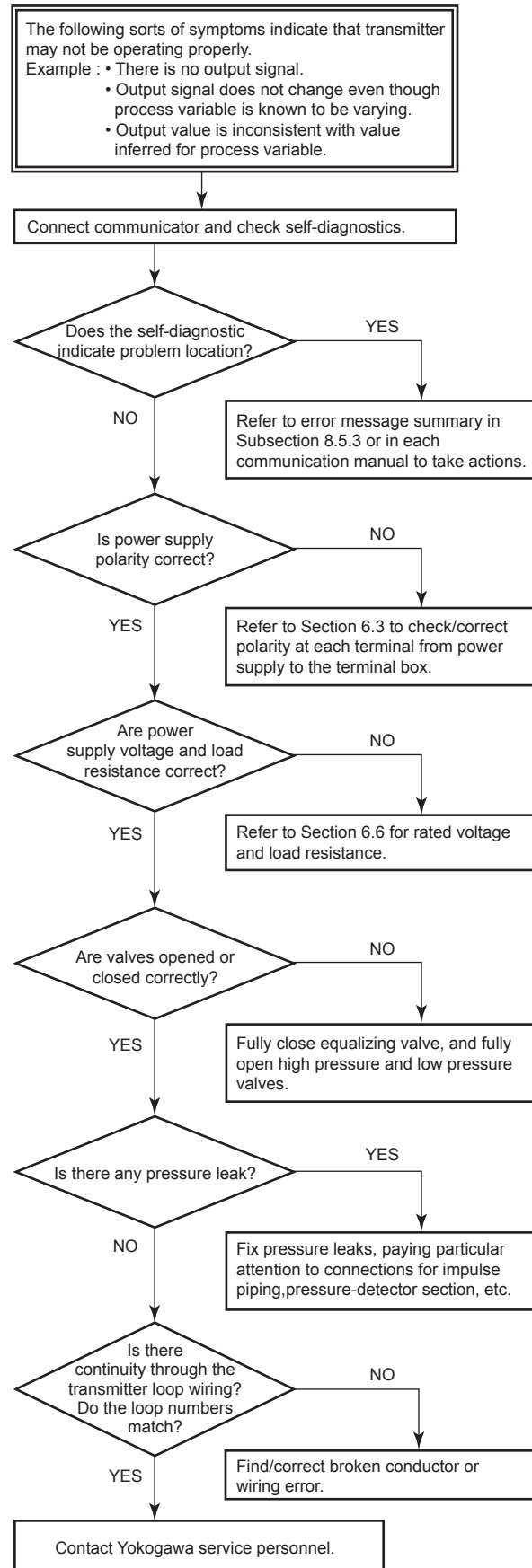
See also each communication manual.



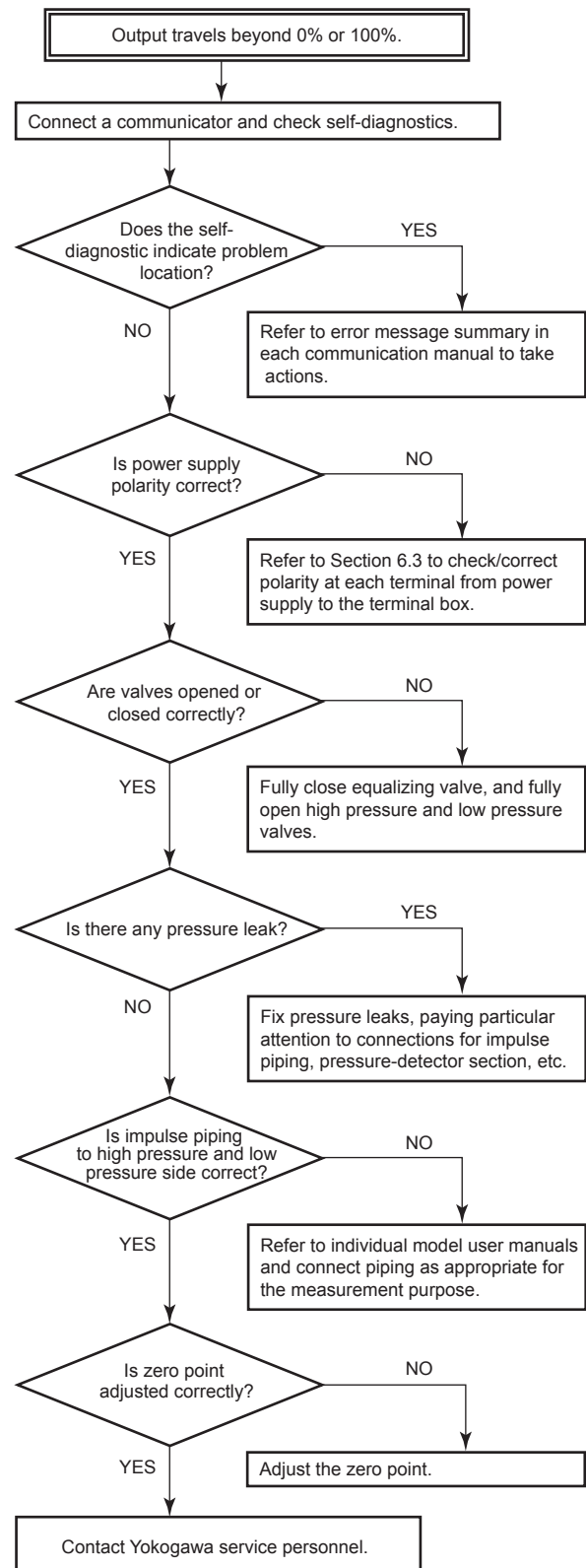
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Figure 8.7 Basic Flow and Self-Diagnostics

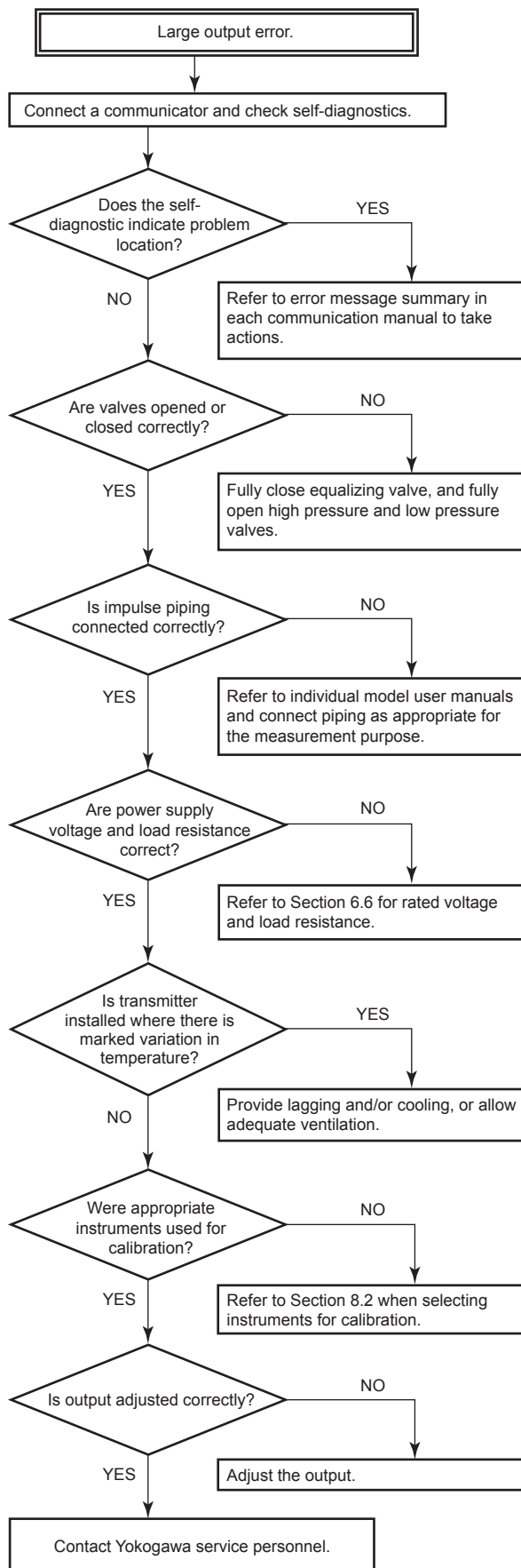
8.5.2 Troubleshooting Flowcharts



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8.5.3 Alarms and Countermeasures

Table 8.3 Alarm Message Summary

Indicator	Cause	Output Operation during Error	Countermeasure
None			
AL. 01 CAP. ERR	Sensor problem. Capsule temperature sensor problem. Capsule EEPROM problem.	Outputs the signal (Hold, High, or Low) set with parameter.	Replace capsule when error keeps appearing even after restart.
AL. 02 AMP. ERR	Amplifier temperature sensor problem. Amplifier EEPROM problem. Amplifier problem.	Outputs the signal (Hold, High, or Low) set with parameter.	Replace amplifier.
AL. 10 PRESS	Input is outside measurement range limit of capsule.	Outputs high range limit value or low range limit value.	Check input or replace capsule when necessary.
AL. 11 ST. PRSS	Static pressure exceeds limit.	Continues to operate and output.	
AL. 12 CAP. TMP	Capsule temperature is outside range (–50 to 130°C).		Use heat insulation or make lagging to keep temperature within range.
AL. 13 AMP. TMP	Amplifier temperature is outside range (–50 to 95°C)		
AL. 30 RANGE	Output is outside upper or lower range limit value.	Outputs high range limit value or low range limit value.	Check input and range setting, and change them as needed.
AL. 31 SP. RNG	Static pressure exceeds specified range.	Continues to operate and output.	
AL. 35 *1 P. HI	Input pressure exceeds specified threshold.		Check input.
AL. 36 *1 P. LO			
AL. 37 *1 SP. HI	Input static pressure exceeds specified threshold.		
AL. 38 *1 SP. LO			
AL. 39 *1 TMP. HI	Detected temperature exceeds specified threshold.		
AL. 40 *1 TMP. LO			
AL. 50 P. LRV	Specified value is outside of setting range.	Holds output immediately before error occurred.	Check setting and change them as needed.
AL. 51 P. URV			
AL. 52 P. SPN			
AL. 53 P. ADJ		Continues to operate and output.	Check input.
AL. 54 SP. RNG		Continues to operate and output holding static pressure in %.	Check setting and change them as needed.
AL. 55 SP. ADJ		Continues to operate and output.	Check input.
AL. 60 SC. CFG	Specified values or settings to define signal characterizer function do not satisfy the condition.		Check setting and change them as needed.
AL. 79 OV. DISP	Displayed value exceeds limit.		

*1: These alarms may appear only when process alarm function is activated.

9. General Specifications

Please refer to the following General Specifications list for the specifications, model, suffix and option codes, and external dimensions of each product.

The General Specifications can be downloaded from the website of Yokogawa.

Website address: <https://www.yokogawa.com/solutions/products-platforms/field-instruments/>

■ General Specifications List

Model	Document Title	Document No.
EJX110A	Differential Pressure Transmitter	GS 01C25B01-01EN
EJX120A	Differential Pressure Transmitter	GS 01C25B03-01EN
EJX130A	Differential Pressure Transmitter	GS 01C25B04-01EN
EJX210A	Flange Mounted Differential Pressure Transmitter	GS 01C25C01-01EN
EJX310A	Absolute Pressure Transmitter	GS 01C25D01-01EN
EJX430A	Gauge Pressure Transmitter	GS 01C25E01-01EN
EJX440A	Gauge Pressure Transmitter	GS 01C25E02-01EN
EJX510A, EJX530A	Absolute and Gauge Pressure Transmitters	GS 01C25F01-01EN
EJX610A, EJX630A	Absolute and Gauge Pressure Transmitters	GS 01C25F05-01EN
EJX118A	Diaphragm Sealed Differential Pressure Transmitter	GS 01C25H01-01EN
EJX118A	Diaphragm Sealed Differential Pressure Transmitter (Inner Diaphragm type)	GS 01C25H01-11EN
EJX438A	Diaphragm Sealed Gauge Pressure Transmitter	GS 01C25J03-01EN
EJX438A	Diaphragm Sealed Gauge Pressure Transmitter (Inner Diaphragm type)	GS 01C25J03-11EN
EJX115A	Low Flow Transmitter	GS 01C25K01-01EN
EJX910A	Multivariable Transmitter	GS 01C25R01-01EN
EJX930A	Multivariable Transmitter	GS 01C25R04-01EN
EJXC50A, EJXC40A, EJAC50E, C20FE, C20FW, C10FR, EJXC80A, EJAC80E, C81FA, C82FA, C81FD, C82FD, C30SW, C30SE, C80FW, C80FE, EJXC81A, EJAC81E, C70SE, C70SW	Diaphragm Seal System	GS 01C25W01-01EN
EJXC40A	Digital Remote Sensor	GS 01C25W05-01EN
EJX110A, EJX130A	Differential Pressure Transmitter High Damping Capsule (Option Code: /HD)	GS 01C25V01-01EN
EJX110A	Differential Pressure Transmitters High Damping Capsule (General) (Option Code: /HD2)	GS 01C25V02-01EN
EJX-A, EJA-E	Explosion Protected Type and Marine Certificate Type	GS 01C25A20-01EN
EJA110E	Differential Pressure Transmitter	GS 01C31B01-01EN
EJA120E	Differential Pressure Transmitter	GS 01C31B03-01EN
EJA130E	Differential Pressure Transmitter	GS 01C31B04-01EN
EJA210E	Flange Mounted Differential Pressure Transmitter	GS 01C31C01-01EN
EJA310E	Absolute Pressure Transmitter	GS 01C31D01-01EN
EJA430E	Gauge Pressure Transmitter	GS 01C31E01-01EN
EJA440E	Gauge Pressure Transmitter	GS 01C31E02-01EN
EJA510E, EJA530E	Absolute Pressure Transmitter and Gauge Pressure Transmitter	GS 01C31F01-01EN
EJA118E	Diaphragm Sealed Differential Pressure Transmitter	GS 01C31H01-01EN
EJA118E	Diaphragm Sealed Differential Pressure Transmitter (Inner Diaphragm type)	GS 01C31H01-11EN
EJA438E	Diaphragm Sealed Gauge Pressure Transmitter	GS 01C31J03-01EN
EJA438E	Diaphragm Sealed Gauge Pressure Transmitter (Inner Diaphragm type)	GS 01C31J03-11EN
EJA115E	Low Flow Transmitter	GS 01C31K01-01EN
EJAC60E, EJA560E	Hygienic Adapter System (Fluidless Type) Hygienic Gauge Pressure Transmitter (Fluidless Type)	GS 01C31Y01-01EN
EJA110E, EJA130E	Differential Pressure Transmitter High Damping Capsule (Option Code: /HD)	GS 01C31V01-01EN
EJA110E	Differential Pressure Transmitters High Damping Capsule (General) (Option Code: /HD2)	GS 01C31V02-01EN

If you cannot find it on our website, please contact YOKOGAWA office.

Revision Information

- Title : Differential Pressure and Pressure Transmitters
EJ□110□, EJ□120□, EJ□130□, EJ□310□, EJ□430□, and EJ□440□
- Manual No. : IM 01C25B01-01E

Edition	Date	Page	Revised Item
1st	Mar. 2004	—	New publication
15th	Oct. 2014	2-2	2.8 • Add information for 1 to 5 V output.
		2-5	2.9.1 • Add information for 1 to 5 V output.
		2-7, 2-8	2.9.2 • Add information for 1 to 5 V output.
		2-9 to 2-11	2.9.3 • Add information for 1 to 5 V output.
		2-13	2.9.4 • Add EPL code. Revise applicable standard. Add information for 1 to 5 V output. Add note for electro static charge. Add electrical connection.
		2-14	2.10 • Add standard for PROFIBUS.
		6-1	6.3 • Add Figure 6.1 Terminal.
		6-2	6.3.3 • Add information for 1 to 5 V output.
		6-3	6.4 • Add information for 1 to 5 V output.
		6-5	6.6 • Add applicability.
		7-6	7.6.1 • Add “10.Device information” in Figure 7.9.
		7-7	7.6.3 • Add note in the figure.
		7-8	7.6 • Add 7.6.6 Pressure LRV/URV Configuration.
		7-9	7.6 • Add 7.6.8 Output Mode Configuration and 7.6.9 Display Out 1 Configuration.
		8-9	8.5.3 • Correct AL31 output operation.
		9-2 to 9-4	9.1 • Add information for 1 to 5 V output.
		9-5	• Add EMC standard for PROFIBUS and EMI specification.
		9-12 to 9-17	9.2 • Add output signal code Q
		9-19	9.3 • Revise the description for SU2 and SF2.
		9-35	9.4 • Add information for 1 to 5 V output.
16th	June 2015	1-3	1.1 • Add (g) and (h).
		2-1, 2-12	Replace name plate figure.
		2-4 to 2-7	NEMA TYPE 4X → Type 4X.
		2-6, 2-7	2.9.2 • Add C22.2 No.61010-2-030
		2-14	2.5 • for Profibus → for Fieldbus
		2-15	2.12 • Add C22.2 No.61010-1, C22.2 No.61010-2-030
		7-7	7.6.3 • Add note for software revision 3.01.
		7-6 to 7-10	Change layout.
		9-5	9.1 • Modify ‘EMC conformity standards.’
		9-26	9.4 • Add Bottom process connection for measurement span code F.
17th	July 2015	2-5, 2-7, 2-9, 2-14	Delete /V1F.
		2-12 to 2-15	2.9.4 • Add SU21. Add a and b.
		9-5	9.1 • Revise descriptions for EMC conformity standards.
		9-6 to 9-17	9.2 • Add material for mounting bracket code D and K.
		9-18	9.2 • Delete #.
		9-19	9.3 • Add SU21.
18th	Sep. 2015	9-3	9.1 • Add condition for Burst Pressure Limits and Maximum Pressure Limits.
		9-21	9.3 • Add /HD.
		9-22 to 9-25	9.4 • Add /HD.
19th	Oct. 2018	1-1	1. • Add S2. • Add descriptions in “Regarding This Manual.”

Edition	Date	Page	Revised Item	
19th	Oct. 2018	1-2 to 1-4	1.1	• Add WARNING marks. Add notes in (c). Add (i) • Delete "1.3 ATEX Documentation."
		2-3	2.8	• Correct current value of dielectric strength test. 10 mA -> 25 mA
		2-4 to 2-5	2.9.1 a.	• Add Control Drawing.
		2-8 to 2-12	2.9.3	• Update ATEX certification. Update Applicable Standards and Type of Protection. (Ex d -> Ex db), Modify "Specific Conditions of Use." Update Name plate for flameproof.
		2-13	2.9.4 a.	• Add description of Enclosure. Modify "Specific Conditions of Use."
		2-14	2.9.4 c.	• Delete c (contents for old /SU2.) Add protection code Gb.
		2-15	2.10	• Update standards.
		2-15	2.11	• Update standards.
		2-16	2.12	• Add "2.12 EU RoHS Directive."
		2-16	2.13	• Delete Altitude.
		4-4	4.5	• Add WARNING. Modify descriptions for rotation.
		7-3	7.2	• Add NOTE.
		8-5	8.4.3	• Add wetted parts material L and W in the table.
		9-3	9.1	• Add wetted parts material W.
		9-5		• Modify description for Housing.
		9-18	9.2	• Add wetted parts material L and W.
		9-19		• Add Table 4.
		9-20 to 9-23	9.3	• Add codes for Local Ex-certificate, Marine certificate, and option codes for Diaphragm Seal System.
20th	July 2019	8-5	8.4.3	• Add the description for optional code HD2.
		9-3	9.1	• Add /HD2 for the description of Burst Pressure Limits and Maximum Pressure Limits.
		9-3	9.1	• Add notes of /HD for Process Temperature Limits.
		9-5	9.1	• Modify the description for Housing. Add Coating of housing.
		9-22	9.3	• Add optional code HD2.
		9-22	9.3	• Add notes for color change option and coating change option.
		9-24 to 9-38	9.4	• Add the note and optional code HD2.
21st	Nov. 2019	Toc-3, 1-1		• Change the description for using the transmitters in Safety Instrumented Systems (SIS) application.
		2-5	2.9.1	• Changed the applicable standard for FM Explosionproof Type.
		7-7	7.6.1	• Add Loop test description, figure and notes.
		7-8	7.6.3	• Add Loop test description and figure.
		7-9		• Add 7.6.4 Loop Test Configuration.
		7-12	7.6.14.1	• Change the figure.
		9-1 to 9-38	9.	• Delete the description for general specifications and add the General Specifications list.
22nd	Apr. 2020	2-6 to 2-9	2.9.2	• Updated the description for CSA Certification.
		2-10, 2-11, 2-13	2.9.3	• Updated the description for ATEX Certification.
		2-16	2.10	• Added some information for EMC Conformity Standards.
		2-18	2.13	• Added (3) and (4).
23rd	July 2021	2-9	2.9.3 a	• Update all the descriptions.
		2-10	2.9.3 b	• Update marking code.
		2-12	2.9.3 c	• Update all the descriptions.
		2-12	2.9.3 (2)	• Delete screw size and marking. Delete WARNING.
		2-13	2.9.3 (3)	• Update tag plates.
		2-13	2.9.4 a	• Update all the descriptions.
		2-17	2.12	• Change EU RoHS Directive.

Edition	Date	Page	Revised Item
24th	Jan. 2023	1-4	1.1 (h) • Add description about importer.
		2-1	2. • Add information about bracket code P and option code /BS.
		2-3	2.9 • Add information to CAUTION.
		2-13	2.9.4 • Add "SU22".
		2-16	• Add "d. IECEx Flameproof Type."
		2-18	2.11 • Add V range for EJ□130□. Add TS and *2 in the table.
		2-19	2.12 • Add USA to the production site.
		3-1	3. • Add note for /BS, /CJ and /CK.
		4-2	4.2 • Add position adjustable bracket to Figure 4.3.
			• Add "4.2.1 How to install the position adjustable bracket"
		5-1	5.1.1 (1) • Add information for /BS.
		6-1	6.1 • dust cap → dust cap(plug)
		6-4	6.4.2 (2) • Add IMPORATNT for flameproof type.
		7-3	7.2 • Add note for /CJ and /CK in IMPORTANT.
		7-6	7.6 • Add note for /CJ and /CK in IMPORTANT.
		8-3	8.4 • Add Allen wrench for ASME.
		8-5	8.4.3 • Add Torque for bolts and nuts code K and H.
		8-6	8.4.5 • Add "Cleaning/replacing the bug screen(Option Code /BS)"
25th	June 2025	2-14	2.9.4 • Change the descriptions for IECEx Intrinsically Safe Ex ic.
		2-19	2.12 • Add UAE to the condition of the RoHS compliant production site.
		6-4	6.4.2 • Revise information of IMPORTANT for requirement of IEC 60079-14.
26th	Sep. 2025	2-6	2.9.2 • Revise Applicable Standard of CSA Intrinsically Safe Type.
27th	Nov. 2025	1-4	1.1 (i) • Change the descriptions. Delete China RoHS.
28th	Jun. 2026	2-3 to 2-18	2.9 • Change the descriptions. Delete detailed explosion_protection requirements.